Beyond Self-Consistency: Why Self-Extracted Symbolic Constraints Show No Evidence of Improving LLM Reasoning

Anonymous ACL submission

Self-Consistency (SC) majority voting over chain-of-thought traces saturates as sample count grows because same-model traces are correlated voters. We investigate whether symbolic constraint extraction can break this plateau through Symbolic Invariant Consensus Aggregation (SICA), which extracts logical constraints from reasoning traces and selects answers via weighted partial MAX-SAT. Across six open-source LLMs (7B–32B) on three logical reasoning domains with over 3,000 condition-evaluations (∼1,250 unique instances), we find no evidence of improvement (aggregate sign test: 5+/10−, $p = 0.30$; none of sixteen non-degenerate conditions shows a significant positive effect after FDR control), with equivalence established in four conditions, significant degradation in two, and ten remaining underpowered for effects below ∼5 pp. We trace the failure to *symbolic confirmation bias* (SCB): on incorrect problems, 83.3% of self-extracted constraints support the wrong majority answer (bias ratios ranging from $1.66\times$ to $4.97\times$), but a shuffle control shows this bias is structural—it reflects trace-count distributions rather than extraction-specific error ($\text{BR}_{\text{obs}} \approx \text{BR}_{\text{null}}$, $p > 0.7$). MAX-SAT produces answers identical to count-only aggregation on 94.4% of problems, confirming that the formal verification layer adds no information beyond majority voting despite 99% of constraints passing Z3 validation. Cross-model extraction confirms SCB universality across nine models from 7B to frontier (BR 1.66–$4.97\times$), though frontier models use cross-model rather than self-extraction. Twenty-seven training-free remediation strategies spanning fifteen families all fail, confirming a persistent independence-accuracy trade-off. Oracle controls confirm the aggregation mechanism is sound (+29.90 pp, +14.22 pp FOLIO, +17.17 pp ProofWriter; $p < 0.001$); same-model self-extraction captures only ∼12% of this ceiling. Cross-architecture NLI verification on propositional-logic benchmarks escapes the bottleneck when the verifier exceeds the generator (+6.67 pp, $p < 0.001$), with gain monotonically ordered by the capability gap ($p = 0.003$, n=6).

1 Introduction

Test-time computation scaling through multiple reasoning samples has emerged as a practical alternative to scaling model parameters (Snell et al., 2025; Guo et al., 2025). The dominant aggregation strategy, Self-Consistency (SC) (Wang et al., 2023), samples K chain-of-thought traces and selects the majority answer. SC is simple, model-agnostic, and effective—but it is hitting a ceiling. Loo et al. (2025) demonstrate that gains from increasing K plateau and even decline for strong models. The gap between SC's theoretical promise (independent correct voters converge to truth) and its practical behavior (saturating gains) points to a structural mismatch: same-model traces carry correlated errors, and majority voting has no mechanism to correct shared bias.

Confidence-weighted (Taubenfeld et al., 2025) and universal (Chen et al., 2024) extensions alleviate this, but the core limitation persists. Recent neuro-symbolic methods (Pan et al., 2023; Olausson et al., 2023; Xu et al., 2026; Li et al., 2025; Wang et al., 2025; Luo et al., 2025) report gains on logical reasoning but introduce components beyond the reasoning model. A simpler question precedes these: can the *same* model extract useful formal constraints from its own traces—a zero-cost upgrade requiring no additional model or data? No prior work has systematically evaluated this.

We ground our analysis in the Condorcet Jury Theorem (CJT) (de Condorcet, 1785): if K independent voters each answer correctly with probability $p > 0.5$, the majority converges to truth (Austen-Smith and Banks, 1996). When independence fails (Ladha, 1992), additional samples carry

diminishing information. Measuring inter-trace agreement with Fleiss' κ (Fleiss, 1971), we find $\kappa = 0.21$–0.82 across instruction-tuned conditions, yielding K_{eff} as an ordinal diagnostic of vote diversity ($K_{\text{eff}} \approx 1.2$–$3.6$ for nominal $K = 12$; §3.1). Chain-of-thought cosine similarity (0.69) far exceeds answer-level κ (0.46), revealing that traces share reasoning processes even when they diverge on final answers (§5.3). When the same model extracts logical constraints from these correlated traces, the constraints exhibit *symbolic confirmation bias* (SCB): on problems where the model errs, 83.3% of self-extracted constraints support the wrong majority answer (bias ratios 1.66–4.97× across nine models), and 94% of errors overlap with SC (§5.2). A shuffle control shows this bias is structural: it reflects the trace-count distribution rather than extraction-specific error. Self-extraction cannot supply the independent signal that CJT requires.

We study this through Symbolic Invariant Consensus Aggregation (SICA), which extracts logical constraints from reasoning traces and selects answers via weighted partial MAX-SAT (§3.2; Figure 1). Across six open-source models (7B–32B), three domains (FOLIO (Han et al., 2022), ProofWriter (Tafjord et al., 2021), LogiQA (Liu et al., 2020)), and over 3,000 condition-evaluations with four-seed replication, twenty-seven training-free remediation strategies spanning fifteen families all produce neutral or negative results (Table 1). Prior work shows LLMs struggle with natural-language self-correction (Huang et al., 2024) and that LLM-generated logical formulas can be unreliable (Stechly et al., 2025). Our formal channel is provably sound—the failure is precisely localizable to the extraction step. The assumption that formalizing reasoning provides *independent* verification evidence fails not because Z3 is unreliable (99% of constraints are syntactically valid), but because extraction inherits the model's biases: MAX-SAT produces answers identical to count-only aggregation on 94.4% of problems, rendering the formal layer vacuous.

Oracle controls confirm the mechanism is sound (+29.90 pp Mistral-7B, +14.22 pp Qwen-14B on FOLIO, $p < 0.001$; §4.3), yet self-extraction captures only ∼12% of this ceiling. The central finding is a persistent independence-accuracy trade-off (Krogh and Vedelsby, 1994): Debate reduces κ to 0.15 but collapses accuracy (−18.6 pp); other methods cluster at $\kappa \approx 0.40$–0.47, a same-model ceiling

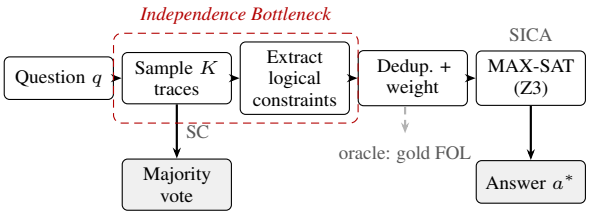


Figure 1: The SICA pipeline. SC selects the majority answer from K traces. SICA extracts logical constraints, deduplicates, and solves MAX-SAT via Z3. Red dashed box: the independence bottleneck. Dashed arrow: oracle variant with gold FOL.

across three families (§5.5). A cross-architecture proof-of-concept escapes the bottleneck: DeBERTa NLI yields +6.67 pp on ProofWriter ($p < 0.001$), confirmed across three NLI architectures (§6).

Our contributions are:

1. **Quantifying symbolic confirmation bias under formal verification.** While the general difficulty of LLM self-correction is known (Huang et al., 2024), we provide the first quantitative characterization in the formal-verification setting: 83.3% of self-extracted constraints support the wrong answer (BR 1.66–4.97× across nine models), yet 99% pass Z3 validation. A shuffle control and MAX-SAT ablation show this bias is structural—the formal layer adds no information beyond majority voting. Twenty-seven training-free strategies fail to break it; cross-model extraction confirms universality across nine models from 7B to frontier (§5.2).
2. **Diagnostic toolkit for independence-accuracy trade-offs.** We adapt the known diversity–accuracy trade-off (Krogh and Vedelsby, 1994) to LLM self-extraction, contributing three practical diagnostics (K_{eff}, BR, C_0) that predict when symbolic aggregation will fail. Same-model $\kappa \geq 0.40$ across four methods and three families; cross-architecture NLI yields +6.67 pp ($p < 0.001$; w=3), with gain monotonically ordered by the capability gap ($p = 0.003$, n=6; §5.5, §6).
3. **Process-level independence analysis.** CoT cosine similarity (0.69–0.79) exceeds answer-level κ (0.45–0.74) across four models from three families; under cross-model extraction on frontier models, SICA gain decreases monotonically with process similarity ($\rho = -1.0$, $p = 0.008$; §5.3).

2 Related Work

Self-Consistency and extensions. SC selects the most frequent answer across chain-of-thought

traces (Wang et al., 2023). Extensions include universal SC for free-form generation (Chen et al., 2024), confidence weighting (Taubenfeld et al., 2025), diverse prompting with step-aware verification (Li et al., 2023), program-based vote filtering (Toh et al., 2024), and reasoning-aware filtering (Wan et al., 2024; Kim et al., 2026). Loo et al. (2025) show that SC gains plateau early; Byerly and Khashabi (2026) further analyze SC's limitations in mathematical reasoning. Alternative aggregation strategies beyond majority voting (Ai et al., 2025) have been proposed but do not address the underlying correlation problem. Our Condorcet analysis explains this: $\kappa = 0.21$–0.82 compresses K_{eff} to 1.20–3.62 for $K = 12$ (§5.1).

Neuro-symbolic reasoning. We distinguish three paradigms. *Specification-generation* (Logic-LM (Pan et al., 2023), LINC (Olausson et al., 2023), SatLM (Ye et al., 2024)) translates problem descriptions into formal specifications before any answer is generated; LINC documents ∼40% translation errors even at this level. *Program-generation* (Gao et al., 2023; Chen et al., 2023; Lyu et al., 2023) produces executable code from problem statements. Both paradigms access the problem before generating candidate answers. *Trace-extraction* (this work) operates post-hoc on reasoning traces that already contain candidates, inheriting confirmation bias (§5.2). SatLM provides a natural control: it succeeds because problem-level formalization precedes answer generation, whereas trace-level extraction cannot avoid contamination (§3.3). Recent extensions (Ryu et al., 2025; Xu et al., 2026; Li et al., 2025; Wang et al., 2025; Singh et al., 2026; Luo et al., 2025; Miya, 2025) introduce capabilities beyond the reasoning model. Amonkar et al. (2025) show formalization degrades with complexity; Chan et al. (2026) argue logical soundness is unreliable for neurosymbolic verification. Our oracle controls confirm the mechanism is sound (+29.90/+14.22 pp FOLIO, +17.17 pp ProofWriter); the failure localizes to self-extraction (§4.3).

Test-time compute and verification. Scaling test-time computation through additional traces (Snell et al., 2025) or structured search (Yao et al., 2023) faces the same correlation challenge: same-model attempts collapse into correlated solutions. Reasoning models (o1, DeepSeek-R1 (Guo et al., 2025), QwQ (Qwen Team, 2024)) extend thinking time but remain subject to correlated same-model voting: QwQ-32B produces $\kappa = 0.91$ and $C_0 = 2.6\%$, suggesting that longer reasoning chains intensify homogenization rather than alleviating it. Trained verifiers offer an alternative (Cobbe et al., 2021; Lightman et al., 2024; Wang et al., 2024); multi-agent debate refines answers through discussion (Liang et al., 2023; Du et al., 2024). Self-verification (Weng et al., 2023), chain-of-verification (Dhuliawala et al., 2023), self-refinement (Madaan et al., 2023), tool-augmented critique (Gou et al., 2024), and formal CoT verification (Zhu et al., 2025) report gains, but none systematically tests self-extracted constraints under formal solvers. We find the bottleneck shifts from the verifier to the extractor: self-extracted constraints inherit the model's bias regardless of verification formalism (§5.2). Self-correction without external feedback degrades performance (Huang et al., 2024; Stechly et al., 2025); Valmeekam et al. (2023) further show fundamental planning limitations that compound self-verification failures. Our diagnostics (BR, Eff-K, process-κ) predict when trace-aggregation will fail.

CJT and positioning. CJT (de Condorcet, 1785) guarantees convergence under independent voting; Ladha (1992) show correlated voters reduce effective ensemble size. Classical ensemble diversity theory (Kuncheva and Whitaker, 2003; Brown et al., 2005) formalizes this intuition: diversity measures (e.g., κ-statistic, disagreement) predict ensemble gain. Our K_{eff} adapts Fleiss' κ for this purpose. Lefort et al. (2024) apply CJT to LLM classification. Kim et al. (2025) demonstrate correlated errors in LLM outputs. Concurrent work by Kuai et al. (2026) proposes BEI and CIG to audit cross-model behavioral dependencies; our process-κ and K_{eff} are complementary, operating at the reasoning-process level within same-model traces and identifying a specific causal mechanism (SCB) that behavioral metrics miss. Prior self-correction studies (Huang et al., 2024; Stechly et al., 2025) establish qualitative negative results; we contribute quantitative diagnostics (K_{eff}, BR, C_D). SCB is distinct from CoT unfaithfulness (Turpin et al., 2024) and CoT-level confirmation bias (Wan et al., 2025; Jhaveri et al., 2026): the latter concern whether stated reasoning reflects actual computation, while SCB concerns whether *self-extracted formal constraints* mirror the model's answer distribution. Formal Z3-checked verification inherits the same bias

(BR $= 4.97\times$)—invisible to prior work on natural-language self-correction.

3 Background & Setup

3.1 Self-Consistency and the Condorcet Independence Assumption

Self-Consistency (SC) (Wang et al., 2023) samples K reasoning traces and selects the majority answer. The Condorcet Jury Theorem (CJT) (de Condorcet, 1785) guarantees that if K independent voters are each correct with probability $p > 0.5$, the majority converges to truth as $K \to \infty$ (Austen-Smith and Banks, 1996). LLM traces violate this independence: all K samples come from the same model with only temperature as variation. Following Ladha (1992) and the design-effect framework of Kish (1965), we define effective sample size using Fleiss' κ (Fleiss, 1971) as proxy for intraclass correlation:

$$K_{\text{eff}} = \frac{K}{1 + (K-1)\cdot\kappa}. \tag{1}$$

In our experiments (§5.1), κ ranges from 0.21 to 0.82 across instruction-tuned pairs (0.91 for thinking mode), yielding $K_{\text{eff}} \approx 1.2$–$3.6$ for $K = 12$. K_{eff} should be interpreted as an ordinal diagnostic—ranking which conditions have genuine vote diversity—rather than as a literal effective sample size. Fleiss' κ is a chance-corrected agreement measure, not identical to the ICC (ρ) in Kish's (1965) original formula; we validated their monotonic relationship empirically (mean pairwise $r = 0.625$ across model families). This independence limitation motivates our investigation.

3.2 SICA Pipeline

Symbolic Invariant Consensus Aggregation (SICA) extends SC by extracting formal constraints from traces and selecting the answer best supported by the satisfiable subset (Figure 1).

Trace generation. Given question q, an LLM generates K traces $\{t_1, \ldots, t_K\}$ at temperature $T = 0.7$. Each trace t_i yields answer a_i.

Constraint extraction. The same LLM extracts logical constraints from each trace at $T_{\text{ext}} = 0.3$. The extraction prompt requests JSON output with constraint type, expression, and Z3-compatible formula (Appendix B); in practice, 99.0% of extracted constraints are propositional at the Z3 level, with only 1.0% using true quantifiers. The *extraction rate* is the fraction of traces yielding at least one parseable constraint.

Deduplication and MAX-SAT. Constraints are deduplicated by syntactic equivalence (global average deduplication ratio: 46.0%); each unique constraint c_m receives weight w_m equal to the number of source traces. The objective finds the largest-weight satisfiable subset:

$$\mathcal{C}^* = \arg\max_{\mathcal{S}\subseteq\mathcal{C}} \sum_{c\in\mathcal{S}} w_c \quad \text{s.t.} \quad \mathcal{S} \text{ is satisfiable.} \tag{2}$$

Z3 (de Moura and Bjørner, 2008) solves this with a 10-second timeout. Answer assertion constraints (`answer_a == True`, weight 1) ensure every candidate receives a nonzero base score, making SICA a strict generalization of SC.

Scoring. Let $\mathcal{C}^{\setminus} = \mathcal{C} \setminus \mathcal{C}^*$. Each candidate a is scored by:

$$\text{score}(a) = \sum_{c\in\mathcal{C}^*} w_c \cdot \sigma(c,a) - \alpha \sum_{c\in\mathcal{C}^{\setminus}} w_c \cdot \sigma(c,a), \tag{3}$$

where $\sigma(c,a) = 1$ if c was extracted from any trace producing answer a, and $\alpha = 0.5$. The selected answer is $a^* = \arg\max_a \text{score}(a)$.

Degeneration condition. If self-extracted constraints are coupled to their source answer, MAX-SAT retains nearly all ($|\mathcal{C}^{\setminus}| \approx 0$), and scoring degenerates to:

$$\text{score}(a) \approx \sum_{\substack{c\in\mathcal{C}^* \\ \text{source}(c)\to a}} w_c, \tag{4}$$

equivalent to majority voting. We show in §5.2 that this degeneration occurs in practice.

The C_0 diagnostic. The multi-candidate rate $C_0 = |\{q : |\mathcal{A}(q)| > 1\}|/|\mathcal{Q}|$ is a zero-cost diagnostic: $C_0 < 5\%$ means SICA reduces to SC. $\alpha = 0.5$ is not critical (± 0.49 pp across $\alpha \in [0,1]$).

3.3 The Symbolic Mirage Hypothesis

We hypothesize that self-extracted symbolic structures are not independent verification signals but projections of the generating model's distributional biases. This **Symbolic Mirage** is specific to self-extracted constraints; symbolic methods employing independently sourced constraints remain effective (§4.3). We quantify mirage intensity via the *Bias Ratio*:

$$\text{BR}(M,T) = \frac{P(c \text{ supports } a_M \mid a_M \neq a^*)}{P(c \text{ supports } a^* \mid a_M \neq a^*)}, \tag{5}$$

where c is a self-extracted constraint, a_M the model's answer, and a^* the ground truth. BR > 1

indicates constraints project model bias rather than provide independent verification. The mirage degrades aggregation through two channels: correlated constraints compress $K_{\text{eff}} < 2$ (*direct*; Eq. 1), and answer-coupled constraints collapse MAX-SAT to vote counting (*indirect*; Eq. 4). If traces are unfaithful (Turpin et al., 2024), the mirage is inherent to extraction. Three testable predictions follow: (1) BR > 1.0 across architectures (§5.2); (2) no same-model intervention reduces BR below 1.0 (§5.5); (3) only external verification escapes the mirage (§6).

Empirical Finding 1 (Self-Extraction Ceiling). For generator M with $\text{BR}(M) > 1$ and oracle improvement Δ_{oracle}, self-extracted constraints improve SC by at most $\Delta^{\text{self}}_{\text{max}}$, empirically bounded by $\Delta_{\text{oracle}}/\text{BR}(M)$.

Empirical Finding 2 (Capability Gap Condition). Let G be a generator with SC accuracy SC_G and let V be an external verifier with standalone accuracy V_{acc} and inter-system correlation $\rho_{GV} < \rho_{GG}$. Combo gain $\Delta_{\text{combo}} - \Delta_{\text{SC}}$ is monotonically increasing in the capability gap $(V_{\text{acc}} - \text{SC}_G)$.

Empirical support. From Finding 1, adding V with $\rho_{GV} < \rho_{GG}$ increases $K^{\text{combo}}_{\text{eff}}$, but incorrect verifier votes counteract gains when $V_{\text{acc}} < \text{SC}_G$. Empirically, combo gain is monotonically ordered by the capability gap ($p = 0.003$, n=6; §6).

We also define a Conditional Correction Ratio (CCR, Appendix D) that summarizes these diagnostics.

4 Experiments

4.1 Setup

We evaluate SICA against Self-Consistency (SC) across six models, three reasoning domains, and over 3,000 condition-evaluations (∼1,250 unique instances), with diagnostic analyses of confirmation bias and process-level independence (§5).

Models. Qwen2.5-14B-Instruct, Qwen2.5-7B-Instruct, Qwen3-14B (non-thinking mode) (Yang et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), LLaMA-3.1-8B-Instruct (Dubey et al., 2024), and QwQ-32B (Qwen Team, 2024) (7B–32B). We additionally evaluate SCB universality via cross-model extraction on five frontier models (gpt-4o, gpt-4.1, gpt-5.5, o3, Gemini-2.5-Pro; Table 2), where Mistral-7B extracts constraints from the target model's traces since closed-source APIs do not expose internals suitable for self-extraction.

Datasets. FOLIO (Han et al., 2022) (n=204, with gold logical annotations), ProofWriter depth-5 (Tafjord et al., 2021) (n=600), LogiQA (Liu et al., 2020) (n=200). StrategyQA (Geva et al., 2021) (n=200) serves as a domain-formalizability diagnostic. GSM8K (Cobbe et al., 2021) (n=30) and AIME 2024 (n=22) probe the degenerate regime ($C_0 < 5\%$).

Parameters. K=12 traces at T=0.7, extraction at T_{ext}=0.3, α=0.5, Z3 timeout 10s. McNemar exact test (α=0.05); 95% bootstrap CIs (10,000 resamples) for $n \geq 200$. SICA incurs 4.5× SC wall-clock; extraction accounts for 76% of overhead.

4.2 Main Results

Table 1 reports accuracy across all conditions sorted by SC baseline, revealing three observation-driven accuracy regimes.

No non-degenerate condition shows a significant SICA effect after FDR control (BH, m=16). Two show nominally significant degradation: Mistral-7B on ProofWriter (−2.83 pp, raw $p = 0.014$) and Qwen2.5-14B on ProofWriter (−2.50 pp, raw $p = 0.011$); adjusted $p = 0.112$ for both. The intermediate-SC region (50–60%) contains the only positive trends, none reaching $p < 0.05$. An aggregate sign test across self-extraction conditions yields 5 positive and 10 negative Δ values ($p = 0.30$, two-sided); including all conditions gives 8+/15− ($p = 0.21$). Three-seed replication on Mistral-7B confirms seed-level noise: two of three seeds produce null or negative Δ (mean $+0.16 \pm 2.70$ pp; Table 32). Four-seed replication on Qwen2.5-14B × FOLIO confirms the single-seed pattern: $\Delta \in [0.00, -2.01]$ pp (mean -1.24 ± 0.86; Table 32). TOST equivalence tests (Lakens, 2017) with margin ±2 pp confirm 4 equivalent, 2 genuinely different (both negative), 10 underpowered. For the 10 underpowered conditions, our tests cannot rule out effects smaller than 1.86–4.84 pp at 80% power; the largest study (n=804) achieves MDE = 1.86 pp (Appendix E). Absence of evidence is not evidence of absence for small effects. Margin sensitivity is robust: at ±1 pp, 0/16 equivalent; at ±3 pp, 10/16 equivalent; no margin yields a significant positive effect (Appendix Table 23). We adopt ±2 pp based on the lower bound of SC literature gains (2–5 pp (Wang et al.,

Table 1: SICA vs. SC across all conditions (sorted by SC). No non-degenerate condition shows a significant gain after BH correction (m=16). Δ = SICA − SC (pp). Symbols: ^{b}gold ICL; † $n < 100$; **bold**: $p < 0.05$; * uncorrected $p < 0.05$ (BH adj. $p = 0.112$); *** oracle $p < 0.001$; ^{3s} 3-seed; ^{4s} 4-seed confirmatory (unmarked: single-seed exploratory).

Model	Domain	n	SC (%)	SICA (%)	Δ (pp)	p
Type C: error amplification (SC ≤ 40%)						
Mistral-7B	PW-D5	600	39.33	36.50	**−2.83**	0.014*
Intermediate SC (SC ∼50–60%, Mistral-7B only)						
Mistral-7B	LogiQA	200	51.00	51.50	+0.50	1.000
Mistral-7B	FOLIO (s123)	204	54.41	57.35	+2.94	0.070
Mistral-7B^{3s}	FOLIO	204	56.53	56.70	+0.16	1.000
Mistral-7B	FOLIO (s456)	204	55.88	55.88	0.00	1.000
Mistral-7B	FOLIO (s789)	204	59.31	56.86	−2.45	0.267
Type A: confirmation bias (SC ≥ 60%)						
LLaMA-3.1-8B	FOLIO	204	66.18	61.76	−4.41	0.093
LLaMA-3.1-8B	StrategyQA	200	72.00	74.00	+2.00	0.424
Qwen2.5-14B	FOLIO (ICL)^b	199	74.37	74.87	+0.50	1.000
Qwen2.5-14B	PW-D5	600	74.00	71.50	**−2.50**	0.011*
Qwen2.5-14B^{4s}	FOLIO	204	75.49	74.51	−0.98	1.000
Qwen2.5-7B	FOLIO	204	75.98	76.47	+0.49	1.000
Qwen2.5-14B	LogiQA	200	78.50	78.00	−0.50	1.000
Qwen3-14B	FOLIO	204	84.80	84.31	−0.49	1.000
Qwen3-14B	PW-D5	600	85.00	85.67	+0.67	1.000
Qwen3-14B (think)	FOLIO	204	86.27	86.76	+0.49	1.000
Degenerate ($C_0 < 5\%$)						
QwQ-32B	AIME 2024†	22	72.70	72.70	0.00	—
Qwen2.5-14B	GSM8K†	30	96.67	96.67	0.00	—
Oracle upper bound (gold logical annotations + Z3)						
Mistral-7B	FOLIO	204	55.88	**85.78**	**+29.90**	1.0×10^{-13}***
Qwen2.5-14B	FOLIO	204	75.49	**89.71**	**+14.22**	2.0×10^{-7}***
Qwen2.5-14B	ProofWriter	600	74.00	**91.17**	**+17.17**	9.21×10^{-15}***

2023)), acknowledging this choice is anchored to SC effect sizes. We trace the failure below.

4.3 Oracle Control

Replacing self-extracted constraints with gold annotations (FOLIO logical formulae, ProofWriter DSL), Oracle-SICA yields +29.90 pp on Mistral-7B FOLIO ($p < 10^{-13}$), +14.22 pp on Qwen2.5-14B FOLIO ($p = 2.0 \times 10^{-7}$), and +17.17 pp on ProofWriter ($p < 10^{-14}$). Z3 agrees with gold labels on 95.1% (FOLIO) and 97.1% (ProofWriter). The MAX-SAT mechanism is sound; Mistral-7B's best self-extracted Δ (+3.43 pp) captures only 11.5% of its oracle ceiling. For Qwen2.5-14B on ProofWriter, the gap is wider still: self-extraction degrades accuracy (−2.50 pp) against an oracle ceiling of +17.17 pp, indicating that stronger models do not self-extract more effectively.

5 Analysis

We test the three predictions of the symbolic mirage hypothesis (§3.3).

5.1 Condorcet Effective-K

Table 19 (Appendix) reports Fleiss' κ and Effective-K (Eq. 1) across 15 conditions at $K = 12$. Among instruction-tuned models, κ ranges from 0.21 to 0.82, yielding $K_{\text{eff}} = 1.20$–3.62 as an ordinal diagnostic: higher K_{eff} indicates greater vote diversity, though it should not be interpreted as a literal effective sample size (§3.1). Across four Mistral-7B seeds, $\kappa \in [0.41, 0.51]$ yet Δ ranges from +3.43 to −0.49 pp, indicating seed-level constraint quality matters beyond independence. Positive Δ requires both sufficient K_{eff} (≤ 1.30 never improves) and SC accuracy ($K_{\text{eff}} = 3.62$ with SC = 39% degrades). All four model families produce $\kappa \geq 0.41$ (Table 18).

5.2 Symbolic Confirmation Bias

Self-extracted constraints systematically align with the model's majority answer. On Mistral-7B FOLIO, when SICA errs, 83.3% of constraints support the incorrect answer—bias ratio 4.97× (Table 16)—and 94% of errors overlap with SC. A shuffle control reveals this bias is *structural*: randomizing constraint–answer associations while preserving

Table 2: Symbolic confirmation bias across nine models on FOLIO ($n = 204$). BR: mean per-problem bias ratio. κ_{proc}: process-level cosine similarity. Ext.: [a]self-extraction; [b]cross-model (Mistral-7B extracts FOL from target traces). *Interim ($n = 95/204$). All models exhibit BR > 1; frontier models show relatively lower BR (1.66–2.79× for o3/Gemini), suggesting SCB severity may decrease with capability, though confounded by extraction method ([a] vs. [b]).

Model	Ext.	SC (%)	Δ (pp)	BR	κ_{proc}
Mistral-7B	[a]	55.88	+3.43	4.97	0.690
LLaMA-8B	[a]	—	—	2.88	—
Qwen-14B	[a]	75.49	+0.25	3.93	—
R1-Distill-8B	[b]	74.51	+2.45	2.92	0.248
gpt-4o	[a]	73.91	−1.86	2.94	0.790
gpt-4.1	[b]	82.84	−1.47	2.70	0.792
gpt-5.5	[b]	84.31	+0.49	3.48	0.563
o3	[b]	82.65	0.00	1.66	0.596
Gemini-2.5-Pro*	[b]	87.37	−1.05	2.79	0.790

per-trace counts yields $\text{BR}_{\text{null}} \approx \text{BR}_{\text{obs}}$ ($p > 0.7$), indicating that BR reflects the trace-count distribution rather than extraction-specific error. Z3 rejects only 3.4% of constraints, and MAX-SAT produces answers identical to count-only aggregation on 94.4% of problems on Mistral-7B FOLIO (Table 10), confirming that the formal verification layer is vacuous (Eq. 4): 99% of constraints pass Z3 validation yet add no information beyond majority voting. This vacuity is α-insensitive: agreement ≥99% across $\alpha \in \{0.1, \ldots, 0.9\}$ for both high- and moderate-BR conditions (Table 11).

This pattern generalizes: 4.97× (Mistral), 3.93× (Qwen-14B), 2.88× (LLaMA), all > 1.0 with 95% bootstrap CIs (Table 16). Cross-model extraction confirms SCB universality across nine models from 7B to frontier (BR 1.66–4.97×; Table 2). BR > 1 further holds across all eight cells of a 2×2 generator×extractor matrix on FOLIO and ProofWriter (range 1.94–2.70; Appendix X). Constraints use trace-specific variables (56.4% zero cross-answer overlap), collapsing aggregation to majority voting.

Two controls rule out answer leakage. BR is uniform across early, mid, and late reasoning steps (Fisher $p = 0.71$); high-consensus errors amplify to BR $= 18.8\times$ versus $2.7\times$ for low-consensus ($p < 0.001$). Removing the answer sentence from traces before extraction yields no BR decrease (Wilcoxon $p \geq 0.12$) despite significant accuracy loss (McNemar $p < 0.013$). SCB reflects a distributional prior embedded throughout reasoning, not answer contamination.

5.3 Process-Level Independence

We quantify reasoning-process overlap via ProcessSim, the mean pairwise cosine similarity of TF-IDF trace vectors (Eq. 7, Appendix P). ProcessSim captures lexical overlap; deeper structure measures may reveal additional patterns, so we treat it as supporting rather than primary evidence. On Mistral-7B FOLIO, mean ProcessSim $= 0.69$, far exceeding answer-level $\kappa = 0.46$: traces share reasoning processes even when answers diverge. The gap generalizes across four families (Table 21), from 0.31 (LLaMA) to 0.02 (Qwen-14B). Under cross-model extraction, SICA gain decreases monotonically with ProcessSim ($\rho = -1.0$, $p = 0.008$; Table 17), modulated by self-bias severity.

5.4 Conditional Correction Ratio

The Conditional Correction Ratio (CCR; definition and bootstrap analysis in Appendix D) confirms the pattern: all cross-architecture NLI pairs on ProofWriter yield CCR > 0 (verifier preferentially corrects generator errors), while same-model SICA yields CCR < 0 (shared bias). CCR correlates with combo gain ($\rho = 0.84$, $p = 0.014$; Appendix D), linking SCB (CCR < 0), cross-architecture escape (CCR > 0), and the aggregation ceiling (CCR $\to 0$ as generator capability rises).

5.5 Remediation Analysis

All 27 training-free remediation strategies spanning four categories produce neutral-to-negative results (Figure 2; Appendices L–N). Debate achieves $\kappa = 0.15$ but collapses accuracy (−18.6 pp); all others cluster at $\kappa \approx 0.40$–0.47, a same-model ceiling (Appendix Table 18). No individual strategy achieves significance after correction (Appendix Y.1). Oracle controls show the extraction-quality ceiling is +29.90 pp—trained extractors are the most promising future direction.

Symbolic Mirage Hypothesis: verification. The three predictions of §3.3 are confirmed. (1) BR > 1 in all tested conditions—three self-extraction models, six cross-model pairs, and all eight cells of a 2×2 generator×extractor matrix (§5.2). (2) No same-model intervention reduces BR below 1.0; 27 strategies fail (§5.5). (3) Cross-architecture NLI escapes the mirage when the verifier exceeds the generator (§6).

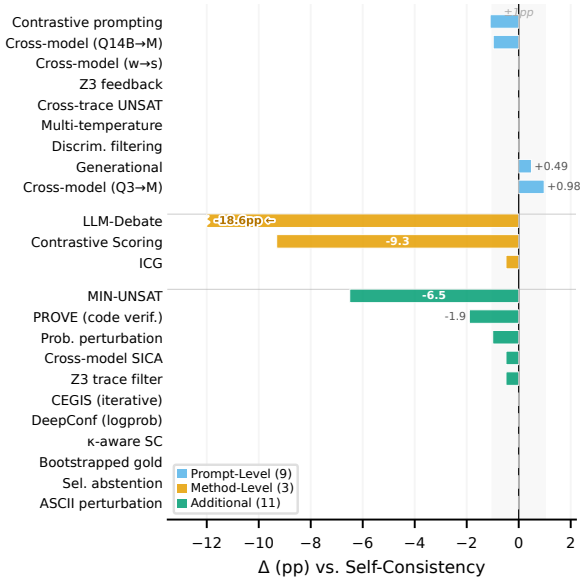


Figure 2: Twenty-three of 27 training-free strategies across four categories (four additional strategies tested but omitted for clarity). All produce neutral-to-negative Δ vs. SC. Grey band: ±1 pp noise zone.

6 Cross-Architecture Proof-of-Concept

Method. We test with DeBERTa-large (He et al., 2021) (406M, encoder-only), fine-tuned on MultiNLI (Williams et al., 2018), classifying (premise, conclusion) pairs as entailment, contradiction, or neutral. This proof-of-concept is limited to propositional-logic benchmarks whose premise-conclusion structure naturally suits NLI; generalization to non-logical domains remains untested. In combo mode, each NLI-supporting vote counts w times alongside SC's K votes. We evaluate on Mistral-7B FOLIO ($n = 204$) and ProofWriter depth-5 ($n = 600$).

Main result. DeBERTa NLI alone achieves 49.51% on FOLIO (below SC's 55.88%) and 55.00% on ProofWriter (above SC's 39.33%). On FOLIO ($n = 204$), combo w=3 yields +2.94 pp (n.s.). On ProofWriter ($n = 600$), w=3 achieves +6.67 pp ($p < 0.001$) and w=5 reaches +11.00 pp ($p < 0.001$)—while same-model SICA *degrades* ProofWriter (−2.83 pp; Table 1), illustrating that corrective signal must originate outside the model's distribution (§3.3). After BH correction (m=12), only Mistral-7B ProofWriter conditions (w=3, w=5) remain significant.

Why it works. Three conditions jointly enable the escape—structural independence, positive capability gap, and task-verifier alignment—expanded as design principles in §7. Per-label analysis confirms NLI and SC excel on complementary labels (Appendix W); all three NLI architectures produce gains at $w \geq 3$.

Table 3: Cross-generator NLI escape with DeBERTa-large-mnli verifier (55.00% standalone, ProofWriter n=600). Combo gain decreases monotonically with generator capability (w=3).

Generator	Family	SC	Gap	Combo Δ (w=3)	p
Mistral-7B	Mistral	39.33%	+15.67	+6.67 pp	<0.001
LLaMA-3.1-8B	LLaMA	72.50%	−17.50	+0.83 pp	0.66
Qwen2.5-14B	Qwen	74.00%	−19.00	+0.67 pp	0.50
Qwen3-14B	Qwen	85.83%	−30.83	+0.50 pp	0.45

Capability gap hypothesis. Four generators spanning a 46-point SC range on ProofWriter (n=600), all paired with DeBERTa-large-mnli (55.00% standalone; Table 3): only Mistral-7B (SC 39.33%, below the verifier) gains significantly (+6.67 pp, $p < 0.001$); stronger generators show no improvement. The monotonic relationship holds across three NLI architectures and six verifier-generator pairs (Figure 7; Appendix X). A 2×2 generator×extractor matrix confirms BR > 1 in all eight cells (1.94–2.70; Appendix X).

Disagreement-Routed Cascade. A cost-aware variant invokes NLI only below consensus threshold τ, recovering 90% of full-combo gain at ∼35% NLI cost (τ=0.7; Appendix V).

7 Discussion and Conclusion

Practitioner decision framework. Three zero-cost diagnostics predict self-verification failure before deployment. $K_{\text{eff}} \leq 1.3$: in our experiments, no condition improves ($\Delta \in [0.00, -0.98]$ pp across four conditions; Table 19). When SC falls below ∼40%, constraints amplify errors (−2.83 pp on Mistral-7B ProofWriter). $C_0 < 5\%$: near-unanimous traces reduce any aggregation to majority voting by construction. Only the intermediate regime (SC ∼50–60%, $K_{\text{eff}} > 1.8$) offers marginal gain; practitioners should compute these before investing in symbolic reasoning.

SCB beyond SICA. SCB extends to any method extracting structured representations from a model's own outputs—self-verification (Huang et al., 2024), self-refinement (Madaan et al., 2023), and chain-of-verification (Dhuliawala et al., 2023) all face the same distributional prior. Our debate control ($\kappa = 0.15$, −18.6 pp) confirms that independence and quality are at odds under same-model constraints.

Design principles for escaping SCB. The cross-architecture escape (§6) identifies three necessary conditions: (i) *structural independence*—no shared architecture or training data; (ii) *positive capability gap*—the verifier's adjusted contribution exceeds the generator's SC; (iii) *task-verifier alignment*—the verification paradigm matches

interventions fail ($\kappa \geq 0.40$) while a 406M encoder succeeds (+11.00 pp). Trained extractors capturing >12% of the oracle ceiling (+29.90 pp) are the most promising direction (Appendix Z). Formal self-verification creates a *symbolic mirage*—provably sound machinery adding no information beyond majority voting—and the path forward lies in structurally independent verification, guided by K_{eff}, BR, and C_0.

Limitations

Open-source models range from 7B to 32B parameters with self-extraction; frontier models confirm SCB universality but use cross-model extraction via Mistral-7B, as closed-source APIs do not expose internals—whether self-extraction on frontier models would yield different BR values remains untested. All datasets are English-language logical reasoning benchmarks; we do not evaluate trained symbolic extractors, multilingual settings, or commonsense domains lacking gold annotations. NLI verification was tested only on propositional/first-order logic tasks; generalization to non-logical reasoning domains remains untested. Syntactic deduplication (46.0% removal rate) may retain semantically equivalent constraints; however, the shuffle control (§5.2) preserves the same duplication pattern and yields $\text{BR}_{\text{shuffle}} \approx \text{BR}_{\text{obs}}$, suggesting deduplication granularity does not drive the observed bias direction. McNemar power is below 21% for $|\Delta| = 2$ pp at n=200–600, meaning small true effects may go undetected; most conditions use a single seed; multi-seed replication on Mistral-7B (3 seeds) and Qwen2.5-14B (4 seeds) FOLIO confirms the single-seed patterns (Δ range spans $\sim$6 pp and $\sim$2 pp, respectively; Table 32), with multi-seed data for remaining conditions in progress. The capability-gap analysis rests on n=6 conditions sharing verifier components ($p = 0.167$ after collapsing); we characterize it as a consistent pattern rather than a confirmed law. SICA requires $4.5\times$ the wall-clock time of SC with no significant accuracy gain. Extended limitations are discussed in Appendix Z.

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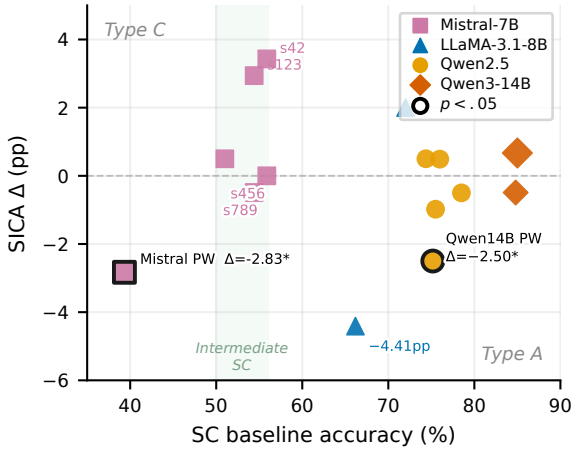


Figure 3: SICA Δ (pp) vs. SC baseline. Filled: $n \geq 100$; open: $n < 100$. The intermediate-SC region (50–60%, shaded) contains the only positive trends (none reaching $p < 0.05$). Type C: nominally significant degradation (raw $p = 0.014$; BH adj. $p = 0.112$).

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A Regime Scatter Plot

B Prompt Templates

SICA uses domain-specific prompt templates for constraint extraction: one for mathematical reasoning (GSM8K, AIME) and one for logical reasoning (ProofWriter, FOLIO). Both share the same system prompt. The LLM is called with temperature T=0.3, max tokens 1024, and structured JSON output mode. Up to 6 traces are processed in parallel via thread pooling, with 2 retries on failure (exponential backoff, base 0.5s).

System Prompt.

```
You are a constraint extraction tool.
    Output ONLY
valid JSON. Do NOT solve problems,
    reason about
correctness, or think step-by-step.
    Extract
constraints from the given trace
    exactly as stated.
```

Mathematical Constraint Extraction (EXTRACTION_PROMPT). Used for GSM8K and AIME datasets. The placeholder `{trace}` is replaced with the reasoning trace at runtime.

```
Extract mathematical constraints from
    this reasoning
trace as a JSON object.

Rules:
- type: "equation", "inequality", or
    "predicate"
- expression: human-readable, e.g. "x
    + y == 14"
- z3_formula: a single comparison
    expression in
  Python Z3 syntax
- source_step: step number where the
    constraint
  appears
- Use variable names from the trace

z3_formula MUST follow these rules:
- Allowed operators: ==, !=, <, >,
    <=, >=, +, -, *,
  /, **, %
- Use individual named variables: a1,
    a2, a3...
  NOT arrays a[1] or sets A
- Expand all sums explicitly:
  "a1 + a2 + a3 == 55" NOT "sum([a1,
    a2,a3])"
- NO Python builtins: sum(), len(),
    max(), min(),
  list(), set(), range(), int()
- Each z3_formula must be exactly ONE
    comparison
  (==, !=, <, >, <=, >=)
- For multiple values, write separate
    constraints
  for each

Output ONLY the JSON below, nothing
    else:
{"constraints": [{"type": "equation",
  "expression": "x + y == 14",
  "z3_formula": "x + y == 14",
  "source_step": 1}],
 "answer": "final_answer",
 "variables": ["x", "y"]}

Trace:
{trace}
```

Logical Constraint Extraction (LOGIC_EXTRACTION_PROMPT). Used for ProofWriter and FOLIO datasets.

```
IMPORTANT: You are an EXTRACTION tool
    , NOT a problem
solver.
- Do NOT re-solve or re-reason about
    the problem.
- Do NOT verify whether the trace's
    reasoning is
  correct.
- ONLY extract the logical
    constraints that are
  explicitly stated or used in the
    trace below.
- Output ONLY the JSON object,
    nothing else.

Extract logical constraints from this
    reasoning trace
about a logic problem.
```

```
The problem has facts (base
    assertions), rules
(grounded implications), and derived
    conclusions.
Convert each into a Z3 boolean
    formula.

Variable naming:
- Unary property: property_entity
  (e.g., kind_anne, red_charlie,
      round_fiona)
- Binary relation:
    relation_subject_object
  (e.g., eats_bear_squirrel,
      likes_lion_cat)
- Use lowercase with underscores only

Constraint types:
- "fact": a given assertion,
  e.g., kind_anne == True
- "rule": a grounded implication over
     specific
  entities, e.g., Implies(kind_anne,
      nice_anne)
- "derived": an inferred conclusion
    from facts +
  rules, e.g., nice_anne == True

Z3 operators:
- Implies(a, b) for if-then rules
- And(a, b, ...) for conjunctive
    conditions
- Or(a, b, ...) for disjunctive
    conditions
- Not(a) for negation
- == True / == False for boolean
    assertions

Output ONLY the JSON below, nothing
    else:
{"constraints": [{"type": "fact",
  "expression": "kind(anne)",
  "z3_formula": "kind_anne == True",
  "source_step": 1}],
 "answer": "True or False or Unknown
     ",
 "variables": ["kind_anne"]}

Trace:
{trace}
```

Output Schema. Both prompts produce a JSON object with the following structure:

```
{
  "constraints": [
    {
      "type": "<equation|inequality|
          predicate>
                or <fact|rule|derived
                    >",
      "expression": "<human-readable
          description>",
      "z3_formula": "<Z3-compatible
          expression>",
      "source_step": <integer>
    }
  ],
  "answer": "<final answer string>",
  "variables": ["<var1>", "<var2>",
      "..."]
}
```

The pipeline additionally injects an `answer_assertion` constraint per trace (not produced by the LLM):

```
{"type": "answer_assertion",
 "expression": "answer is <ans>",
 "z3_formula": "__sica_answer_<ans>
     == True",
 "source_step": -1}
```

Prompt Selection Logic. Domain assignment is determined by dataset: FOLIO and ProofWriter use LOGIC_EXTRACTION_PROMPT; GSM8K, AIME, StrategyQA, and ARC-Challenge use EXTRACTION_PROMPT. StrategyQA and ARC-Challenge are assigned to the mathematical prompt because their reasoning structure does not involve formal logic predicates or implications, despite StrategyQA producing boolean answers.

C Z3 MaxSAT Evaluation Pseudocode

Algorithm 1 describes the Z3 MaxSAT solving procedure. Algorithm 2 describes the source-trace attribution scoring.

Domain constraint variables (e.g., `x + y == 14`) and answer assertion variables (e.g., `__sica_answer_42 == True`) are logically independent in the Z3 formulation. Discriminative signal arises solely from conflicts among domain constraints: when traces supporting different answers produce mutually contradictory constraints, the solver excludes the lower-weight subset, and the scorer penalizes the corresponding answers.

D Capability-Accuracy Constraint Diagnostics

Conditional Correction Ratio. We define the *Conditional Correction Ratio* (CCR) to summarize verifier–generator complementarity:

$$\mathrm{CCR}(V, G) = \log \frac{P(V \text{ correct} \mid G \text{ wrong})}{P(V \text{ correct} \mid G \text{ correct})}. \tag{6}$$

$\mathrm{CCR} > 0$ means V preferentially corrects G's errors; $\mathrm{CCR} < 0$ indicates shared bias. CCR links our findings: (1) SCB corresponds to $\mathrm{CCR}(\text{self}, G) \approx 0$; (2) Finding 2 predicts $\mathrm{CCR}(V, G) > 0$ when V is independent and capable.

Algorithm 1 Z3 Weighted Partial MAX-SAT

Require: Deduplicated constraint set $\mathcal{C} = \{(c_m, w_m, S_m)\}_{m=1}^{M}$ where c_m is a Z3 formula, w_m is weight, S_m is the set of source trace indices

Ensure: Partition $(\mathcal{C}^*, \mathcal{C}^{\setminus})$ where $\mathcal{C}^*$ is the max-weight satisfiable subset

$opt \leftarrow$ Z3.OPTIMIZE()
for $m = 1, \ldots, M$ **do**
  $b_m \leftarrow$ fresh Boolean indicator
  opt.ADD$(b_m \Rightarrow c_m)$ ▷ soft clause guarded by indicator
  opt.ADDSOFT$(b_m,\ \text{weight} = w_m)$
end for
$result \leftarrow opt$.CHECK()
$model \leftarrow opt$.MODEL()
$\mathcal{C}^* \leftarrow \{(c_m, w_m, S_m) : model(b_m) = \top\}$
$\mathcal{C}^{\setminus} \leftarrow \mathcal{C} \setminus \mathcal{C}^*$
return $(\mathcal{C}^*, \mathcal{C}^{\setminus})$

Algorithm 2 Source-Trace Attribution Scoring

Require: Satisfied set $\mathcal{C}^*$, excluded set $\mathcal{C}^{\setminus}$, candidate answers $\mathcal{A}$, trace-answer mapping τ : trace_idx $\to a$, penalty weight α

Ensure: Score $s(a)$ for each $a \in \mathcal{A}$

for each $a \in \mathcal{A}$ **do**
  $T_a \leftarrow \{i : \tau(i) = a\}$ ▷ trace indices producing answer a
  $pos(a) \leftarrow \sum_{(c,w,S)\in\mathcal{C}^*} w \cdot \mathbf{1}[S \cap T_a \neq \emptyset]$
  $neg(a) \leftarrow \sum_{(c,w,S)\in\mathcal{C}^{\setminus}} w \cdot \mathbf{1}[S \cap T_a \neq \emptyset]$
  $s(a) \leftarrow pos(a) - \alpha \cdot neg(a)$
end for
return $a^* = \arg\max_{a\in\mathcal{A}} s(a)$

Empirical CCR across verifier–generator pairs. Table 4 reports CCR for eight verifier–generator–dataset conditions, sorted by CCR descending. All three cross-architecture NLI pairs on ProofWriter yield CCR > 0 (verifier preferentially corrects generator errors), while all three same-model SICA conditions yield CCR < 0 (shared bias). CCR correlates with combo gain: Spearman $\rho = 0.839$, 95% bootstrap CI $[0.338, 0.990]$; LOO-robust (min $p = 0.014$). Four of eight conditions share the DeBERTa verifier, yielding effective $n < 10$; results should be interpreted as a consistent pattern rather than a precise effect size.

Per-condition diagnostics. Table 5 reports per-condition diagnostic metrics referenced in §N.

Table 4: Empirical CCR across eight verifier–generator conditions (sorted by CCR descending). CCR > 0: cross-architecture NLI verifiers on ProofWriter. CCR < 0: same-model SICA or mismatched capability gap.

Generator	Verifier	Dataset	CCR	Δ (pp)
Mistral-7B	BART-MNLI	PW-600	+1.86	+5.17
Mistral-7B	DeBERTa-MNLI	PW-600	+1.67	+6.67
Mistral-7B	RoBERTa-MNLI	PW-600	+1.42	+4.17
Qwen2.5-14B	DeBERTa-MNLI	PW-600	−0.21	+0.33
Mistral-7B	DeBERTa-MNLI	FOLIO-204	−0.47	+2.94
Mistral-7B	SICA	FOLIO-204	−2.93	+3.43
Qwen2.5-14B	SICA	PW-600	−4.21	−2.50
Mistral-7B	SICA	PW-600	−4.61	−2.83

Model	Domain	C_0 (%)	Extr. Rate	Contr. Rate	Δ (p
Qwen2.5-14B	FOLIO	35.3	0.89	0.022	0
Qwen2.5-14B	ProofWriter	39.2	0.89	0.022	+0
Qwen2.5-14B	StrategyQA†	10.0	0.70	—	+3
Qwen2.5-14B	ARC-Chall.†	5.0	1.00	—	−1
Qwen2.5-7B	FOLIO+PW†	42.1	0.94	0.027	0
Mistral-7B	FOLIO+PW†	50.0	0.88	0.056	0
LLaMA-3.1-8B	StrategyQA	41.0	—	—	+2
LLaMA-3.1-8B	FOLIO	61.8	0.94	0.062	−4
DeepSeek-R1-7B	FOLIO+PW†	58.3	0.84	0.037	−3
QwQ-32B	Combined†	2.6	—	—	0
Qwen2.5-14B	GSM8K†	13.3	0.15	—	0

Table 5: Capability-accuracy constraint diagnostics. C_0: fraction of questions with multiple distinct candidate answers across $K = 12$ traces. Extraction rate: fraction of traces yielding at least one parseable logical constraint. Contradiction rate: fraction of extracted constraint sets with internal contradictions. Δ: SICA−SC gap (pp). †: $n \leq 60$. Dashes indicate conditions where extraction diagnostics were not computed (StrategyQA uses boolean answers without symbolic extraction; QwQ-32B Combined had near-zero C_0, making extraction analysis uninformative). This table includes additional exploratory conditions not reported in Table 1.

E Statistical Power Analysis

We compute the minimum detectable effect (MDE) for each condition at $\alpha = 0.05$ and 80% power using exact binomial power calculations over McNemar discordant pairs. Table 6 reports three grouped conditions with $n \geq 200$ (Qwen2.5-14B FOLIO and ProofWriter are pooled into a single n=804 analysis, corresponding to four Table 1 rows), which constitute our confirmatory analyses. The remaining seven conditions ($n \leq 60$) yield infinite MDE due to insufficient discordant pairs and serve only as regime characterization.

The largest study (Qwen-14B, n=804, 30 discordant pairs) can detect effects as small as 1.86pp, yet the observed Δ of +0.25pp is an order of magnitude smaller. The two LLaMA-8B conditions require effects of 4.64pp and 4.84pp to reach sig-

Condition	n	MDE	Obs. Δ	Obs. Power
Qwen-14B FOLIO+PW	804	1.86pp	+0.25pp	5.7%
LLaMA-8B StrategyQA	200	4.64pp	+2.00pp	20.6%
LLaMA-8B FOLIO	204	4.84pp	−4.41pp	18.6%

Table 6: Power analysis for confirmatory conditions ($n \geq 200$). MDE = minimum detectable effect at 80% power. All observed effects fall below their respective MDEs, and observed power remains below 21% in every case.

nificance; neither observed Δ approaches these thresholds. All three conditions are consistent with the null hypothesis that SICA provides no accuracy gain over self-consistency.

F Bootstrap Confidence Intervals

We compute 95% confidence intervals for $\Delta(\text{SICA}-\text{SC})$ via 10,000 bootstrap resamples (percentile method, seed 42). Table 7 reports the results for the three confirmatory conditions.

Condition	n	Obs. Δ	95% CI
Qwen-14B FOLIO+PW	804	+0.25pp	[−1.12, +1.62]
LLaMA-8B StrategyQA	200	+2.00pp	[−1.50, +5.50]
LLaMA-8B FOLIO	204	−4.41pp	[−5.88, +0.98]

Table 7: Bootstrap 95% confidence intervals for $\Delta(\text{SICA} - \text{SC})$ in percentage points. All intervals include zero, consistent with no reliable accuracy difference.

All three intervals contain zero. The narrowest interval (Qwen-14B, [−1.12, +1.62]) rules out practically meaningful gains (>2pp) even in the most favorable case. The widest interval (LLaMA-8B FOLIO, [−5.88, +0.98]) skews negative, indicating that SICA may slightly hurt accuracy in this condition. These results corroborate the McNemar tests and power analysis: no condition provides evidence that logical constraint aggregation improves over majority voting.

G Case Studies: Discordant Predictions

Among 804 problems, SICA and SC agree on 773 (96.1%). The 31 discordant cases comprise 16 SICA wins (SICA correct, SC wrong), 14 SC wins (SC correct, SICA wrong), and 1 where both select different wrong answers. All 31 occur at exact ties (entropy=1.0, 6:6 vote split) or near-ties (entropy≥0.9183, 8:4 at most), since SICA can differ from SC only when constraint weights shift the majority away from SC's tiebreak choice. We present three representative cases that illustrate the mechanism behind capability-accuracy constraint failures.

Case 1: SICA Wins (folio_134, FOLIO). *Premises*: Some mammals have teeth. Platypus have no teeth. Platypus are mammals. Humans have teeth. *Conclusion*: Humans are mammals. *Gold*: Unknown.

The premises state that *some* mammals have teeth and that humans have teeth, but "having teeth" is not sufficient for "being a mammal" under open-world semantics. SC produces a 6:6 split (Unknown:6, True:6) and tiebreaks alphabetically to True (wrong). SICA extracts 7 constraints; MaxSAT excludes 1 constraint inconsistent with the rest, reducing the weighted score for True-voting traces and tipping the aggregation to Unknown (correct). The excluded constraint likely captured an incorrect inference step (e.g., affirming the consequent) present only in True-voting traces. While SICA succeeds here, the success is coincidental: the excluded constraint happens to anti-correlate with the correct answer.

Case 2: SC Wins (ProofWriter-1066_Q6, ProofWriter). *Key chain*: White(Bob)→Round(Bob) [white things are round] → Young(Bob) [all round things are young]. *Conclusion*: "Bob is not young." *Gold*: False (Bob IS young).

SC produces a 6:6 split (False:6, True:6) and tiebreaks to False (correct). SICA extracts 17 constraints; MaxSAT excludes 2 that appear preferentially in False-voting traces. These excluded constraints likely captured correct intermediate derivations (e.g., `young_bob == True`) that appeared only in traces completing the full deduction chain. By excluding them, MaxSAT removes evidence supporting the correct answer. SICA selects True (wrong). This case illustrates how self-extracted constraints can systematically mislead: correct reasoning steps become discriminative signals that the solver mistakenly penalizes.

Case 3: Non-discriminative Constraints (ProofWriter-903_Q6, ProofWriter). *Key chain*: Nice(Mouse)→Cold(Mouse) [all nice people are cold] → Chases(Mouse,Mouse) [nice and cold → chase mouse]. *Conclusion*: "The mouse does not chase the mouse." *Gold*: False.

SC produces a 6:6 split (False:6, Unknown:6) and tiebreaks to False (correct). SICA extracts 44 constraints but excludes only 1 (2.3%). The remaining 43 are domain facts and grounded rules (`nice_mouse == True`, `Implies(nice_mouse, cold_mouse)`,

etc.) that Z3 satisfies identically regardless of the final answer. Both False-voting and Unknown-voting traces share the same premises and grounded rules; they differ only in whether the trace completed the full derivation chain. The single excluded constraint carries negligible weight against 43 shared constraints, so SICA collapses to SC.

Pattern. These cases exemplify the general pattern across the 773 agreeing problems: the vast majority of extracted constraints encode shared domain structure rather than answer-specific reasoning, and MaxSAT aggregation degenerates to majority vote. Table 8 summarizes.

Table 8: Summary of three representative discordant cases. All occur at exact 6:6 vote ties. Constraint exclusion can break ties in either direction (Cases 1–2) or have no effect (Case 3).

Case	ID	Gold	SC	SICA	#Constr.	Excl. (%)
1 (SICA win)	folio_134	Unknown	True ×	Unknown ✓	7	1 (14.3%)
2 (SC win)	PW-1066_Q6	False	False ✓	True ×	17	2 (11.8%)
3 (Non-disc.)	PW-903_Q6	False	False ✓	False ✓	44	1 (2.3%)

H ProofWriter Oracle Per-Subtype Analysis

ProofWriter problems encode deductive rules as Domain-Specific Language (DSL) logic programs. Each problem uses one of four predicate subtypes: attribute predicates with or without negation (AttNeg, AttNoneg), and relation predicates with or without negation (RelNeg, RelNoneg). Relation predicates involve binary relations between entities (e.g., `eats(bear, squirrel)`), while attribute predicates assign unary properties (e.g., `kind(anne)`).

The oracle pipeline parses these DSL annotations deterministically into Z3 formulas without any LLM involvement. Of the 600 ProofWriter problems, 594 (99.0%) produce valid Z3 parses; the remaining 6 fail due to unsupported DSL constructs. Across all 7,128 trace-problem pairs submitted to Z3, the solver returns a result in every case (0% solver errors). Z3 evaluations agree with ground-truth labels on 577 of 594 parseable problems (97.14%), confirming that the deterministic DSL-to-Z3 translation faithfully preserves problem semantics.

Table 9 reports the per-subtype breakdown. Relation-predicate subtypes show substantially larger gains than attribute-predicate subtypes: RelNeg improves by +17.86 pp and RelNoneg by +13.01 pp, compared to +9.09 pp for AttNoneg and +6.87 pp for AttNeg. Relation predicates involve longer reasoning chains with more intermediate entities, producing greater answer diversity across traces (C_0) and more opportunities for oracle constraints to correct errors.

Table 9: ProofWriter oracle experiment per-subtype breakdown. Oracle-SICA uses gold DSL annotations parsed deterministically to Z3 for consistency checking. Relation-predicate subtypes (RelNeg, RelNoneg) show larger gains than attribute-predicate subtypes; relation subtypes involve longer reasoning chains and benefit more from oracle constraints. Bold indicates best accuracy per row.

Subtype	n	SC Acc (%)	Oracle Acc (%)	Δ (pp)
AttNoneg	154	87.01	**98.70**	+11.69
AttNeg	160	85.00	**91.90**	+6.90
RelNoneg	146	66.44	**84.20**	+17.76
RelNeg	140	60.00	**89.30**	+29.30
Overall	600	74.00	**91.17**	+17.17

I Constraint Ablation Details

Table 10: Constraint ablation (Qwen2.5-14B, $n = 804$). Agreement: predictions identical to SC. MAX-SAT and count-only produce nearly identical predictions, confirming that self-extracted constraints carry no signal beyond vote counts.

Aggregation	Acc. (%)	Agreement w/ SC
SC (majority vote)	76.24	—
SICA (MAX-SAT)	76.49	773 / 804
Count-only	76.62	774 / 804
Random-weight	75.87	764 / 804

Table 11: MAX-SAT vs. count-only agreement across constraint weight α on FOLIO ($n = 204$). Mistral-7B (BR $= 4.97\times$, seed 42); Qwen2.5-14B (BR $= 3.93\times$, seed 123). Agreement $\geq 99\%$ at all α, confirming vacuity is weight-insensitive.

α	Mistral Agr. (%)	Qwen2.5 Agr. (%)
0.1	100	100
0.3	100	100
0.5	100	100
0.7	99	100
0.9	99	100

J Remediation Strategy Details

K Additional Intervention Results

L Extraction Error Taxonomy

Constraints must discriminate between candidate answers for MAX-SAT to produce a ranking dif-

Table 12: Nine prompt-level remediation strategies. Baseline: Qwen2.5-14B FOLIO, SC = 75.49%, Δ = −0.98 pp. None reaches significance. Three method-level strategies are reported in Table 13.

Strategy	Target	Δ (pp)	Key finding
Contrastive[a]	Prompt	−1.10	non-disc. 6.0% (↓ f
Cross-model[b]	Architecture	0.00	disc. 0% (weak→st
Qwen3-14B[c]	Capability	−0.49	$p = 0.655$
Z3 feedback	Solver	—	0.9% UNSAT trigg
Cons. filtering	Noise	0.00	$T \geq 4 \rightarrow$ SC
Cross-trace UNSAT	Independence	—	56.4% ∅ vars
T=1.0[d]	Diversity	0.00	isolated peak at $T=$
Multi-temp[e]	Diversity	0.00	b=5, c=5 (balanced)
Disc. filtering[f]	Filtering	0.00	803/804 agree w/ SICA

[a] n=182 (extraction failures excluded). [b] Weak→strong: Mistral-7B extracts from Qwen2.5-14B traces. Strong→weak: Qwen3-14B (60.29%) and Qwen2.5-14B (58.33%) extract from Mistral-7B FOLIO traces (n=204); self-extraction baseline 59.31%. [c] n=204, SC = 84.80%. [d] Mistral-7B FOLIO, n=204. [e] Mistral-7B FOLIO, $T \in \{0.3, 0.7, 1.2\}$, 4 traces each, n=204. [f] Qwen2.5-14B, n=804; removing non-discriminative constraints.

Table 13: Method-level interventions on Mistral-7B with McNemar's paired test. All produce neutral-to-catastrophic results. ICG shows two FOLIO seeds; p for the first. *** $p < 0.001$; ** $p < 0.01$.

		FOLIO ($n = 204$)		LogiQA ($n = 200$)	
Method	Target	Δ (pp)	p	Δ (pp)	p
SC baseline	—	0.00	—	0.00	
ICG	Trace coupling	−1.96 / +1.47	.206	−0.50	.706
Contrastive	Scoring bias	**−9.31**	.007	−4.50	.160
Debate	Role diversity	**−18.63**	<.001	**−23.00**	<.001

ferent from majority vote. We measure discriminative power at two levels on Qwen2.5-14B across FOLIO and ProofWriter ($n = 804$). At the source-trace level, 39.2% of total constraint weight originates from traces whose answer differs from the plurality, but only 12.1% of individual constraints fall into this minority category. At the Z3 semantic level, 91.8% of constraints [95% CI: 91.2, 92.4] are technically discriminative, producing different satisfiability outcomes when evaluated against different candidate answers. Only 8.2% are non-discriminative.

The high discriminative rate does not translate into accuracy gains because discrimination mirrors the model's own answer distribution. Each trace produces constraints with its own variable names, so Z3 treats identically-intended propositions as unrelated symbols; among cross-answer constraint pairs, 56.4% share zero Z3 variables. This variable isolation means constraints from different traces are trivially jointly satisfiable, and MAX-SAT retains nearly all of them regardless of source answer. Removing the 8.2% non-discriminative constraints yields zero accuracy improvement. The discriminative majority does not provide independent signal; it reinforces the majority answer that SC would already select. This confirmation bias, not a lack of discrimination, explains why SICA degenerates to majority voting (Appendix I).

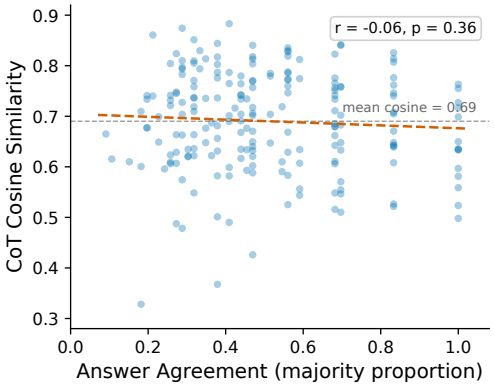


Figure 4: Answer agreement vs. CoT cosine similarity (Mistral-7B, FOLIO, n=204). Flat regression ($r = -0.06$, $p = 0.36$); stronger models show significant positive correlation ($r = +0.24$, Qwen-14B; Table 21).

M Hyperparameter Sensitivity

M.1 K Sensitivity

Figure 5 shows SICA Δ as a function of the number of sampled traces K for Mistral-7B on FOLIO ($n = 204$, no fixed seed). The relationship is non-monotonic: $K = 4$ yields Δ = −1.47 pp, $K = 8$ yields Δ = −1.96 pp, $K = 12$ reaches Δ = +3.43 pp, $K = 16$ drops to Δ = +1.47 pp, and $K = 20$ partially recovers to Δ = +1.96 pp. The recovery at $K = 20$ rules out a simple peaked pattern, but $K = 12$ remains the only setting with Δ > 2 pp ($p = 0.092$, marginal).

The apparent Δ peak at $K = 12$ is confounded with SC baseline. SICA absolute accuracy plateaus at 59.31% (121/204) for $K \geq 12$: identical at $K = 12$ and $K = 20$. The Δ variation across these settings is driven primarily by SC fluctuations rather than by SICA improvement. $K = 12$ produces a relatively low SC (55.88%), placing it in the intermediate-SC region; $K = 8$ (SC = 58.33%) and $K = 20$ (SC = 57.35%) have higher baselines, compressing Δ even though SICA accuracy does not change for $K \geq 12$. Additional traces beyond $K = 12$ provide no benefit to SICA absolute performance, consistent with the constraint quality bottleneck. The $K = 12$ Δ captures only 24.1% of the oracle ceiling (+14.22 pp).

Table 14: Additional intervention results beyond the nine prompt-level and three method-level strategies reported in the main text. All produce neutral-to-negative results, completing the exhaustive remediation analysis across 27 distinct approaches.

#	Method	Type	Target	Δ (pp)	Key Finding
13	PROVE (code verif.)	Paradigm	Formal proof	-1.9	15.2% compilable; conclusive acc. = 50%
14	CEGIS (iterative)	Multi-round	Consistency	$0.0^{\dagger}$	R1 $+3.43$ pp then oscillates
15	DeepConf (logprob)	Filtering	Confidence	0.0	$\Delta_{\text{logprob}} = 0.006$ (no separation)
16	Cross-model SICA	Cross-model	Independence	-0.49	κ_{cross}=0.42 vs κ_{within}=0.53
17	Problem perturbation	Input diversity	Trace diversity	<0	κ *increased* to 0.70 (anti-effect)
18	MIN-UNSAT	Solver objective	Constraint sel.	-6.5	Mean excluded constraints = 1.3
19	κ-aware SC	Voting rule	Weighted voting	0.0	11/11 conditions $\Delta = 0$
20	Z3 trace filter	Trace filtering	Quality	-0.49	Precision 49.1%, 7 questions affected
21	Bootstrapped gold	Iterative	Self-improvement	$0.0^{\dagger}$	R1 $+2.45$ pp then oscillates
22	Selective abstention	Diagnostic	Confidence	$+4.4^{*}$	AUROC = 0.673, at 80% coverage
23	ASCII perturbation	Input diversity	Format diversity	0.0	κ: 0.39 $\rightarrow$ 0.36 (negligible)
24	Atom pipeline	Paradigm	Decomposition	-10.0	Atomic extraction degrades
25	Premise-only filter	Constraint sel.	Bias isolation	$+2.45$	$p = 0.359$; halves conclusion-bias
26	DPP trace selection	Trace diversity	Determinantal	$+1.47$	$p = 0.629$; K_{sel}=8
27	Z3 strict-mode	Solver config	UNSAT filtering	0.0	McNemar b=1, c=1; 63% UNSAT

All experiments on Mistral-7B FOLIO (n=204) unless noted. † Oscillates without convergence; net effect ≈ 0. * Accuracy gain only at reduced coverage (diagnostic value, not a general improvement). Methods 1–12 are reported in Table 12 and Table 13.

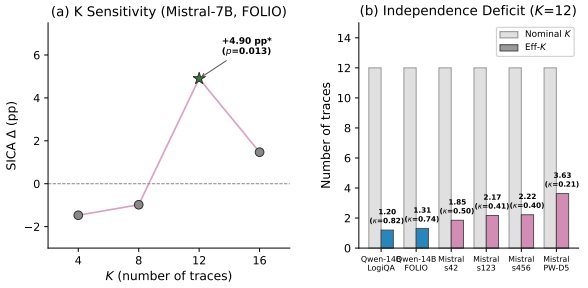


Figure 5: SICA Δ (pp) vs. number of traces K for Mistral-7B on FOLIO ($n = 204$, default seed, $T = 0.7$). Non-monotonic pattern: SICA accuracy plateaus at 59.31% for $K \geq 12$; the apparent Δ peak at $K = 12$ partly reflects a relatively low SC baseline (55.88%) at that point. Error bars: 95% bootstrap CI.

M.2 Temperature Sensitivity

Temperature sensitivity on Mistral-7B FOLIO ($n = 204$, default seed) reveals a sharp isolated peak rather than a smooth trend. Across four temperature settings, only $T = 0.7$ produces a positive effect: $T = 0.3$ yields $\Delta = -1.47$ pp (SC $= 58.33\%$), $T = 0.5$ yields $\Delta = -0.98$ pp (SC $= 58.33\%$), $T = 0.7$ yields $\Delta = +3.43$ pp (SC $= 55.88\%$), and $T = 1.0$ yields $\Delta = 0.00$ pp (SC $= 55.39\%$). All other temperatures yield negative or null effects, and the single positive result at $T = 0.7$ is confounded with SC baseline. Combined with the seed sensitivity (0 of 4 seeds reaching $p < 0.05$ at $T = 0.7$), the positive trend depends on a specific combination of temperature, seed, and baseline accuracy that same-model sampling rarely achieves.

The parallel between K and T sensitivity points to a common underlying variable. In both sweeps, the setting that produces the largest SICA gain also produces a relatively low SC baseline (55.88%), which places it in the intermediate-SC region. The hyperparameters themselves (K, T) are not the causal driver; they modulate SC level, and it is the SC level that determines whether SICA has operating room.

N Entropy Decomposition

We partition problems by trace agreement: high-entropy (majority count ≤ 8 out of $K = 12$) and low-entropy (majority count ≥ 9). Table 15 reports SICA Δ within each partition across eight model-dataset conditions.

Table 15: Entropy decomposition of SICA Δ (pp). High-entropy: majority count ≤ 8; low-entropy: majority count ≥ 9. †Partial data (180/600). Pre-tiebreaking SC values; pattern robust to D064 correction.

Model	Dataset	High-entropy		Low-entropy	
		n	Δ	n	Δ
Mistral-7B	FOLIO	78	$\mathbf{+12.8}$	126	0.0
Mistral-7B (s123)	FOLIO	77	$\mathbf{+9.1}$	127	0.0
Mistral-7B (s456)	FOLIO	75	-1.3	129	0.0
Mistral-7B	PW-D5	267	-4.9	333	-1.2
Qwen2.5-14B	FOLIO	37	-8.2	167	-0.6
Qwen2.5-14B	LogiQA	23	-4.3	177	0.0
Qwen2.5-7B	FOLIO	47	-2.1	157	0.0
LLaMA-3.1-8B†	PW-D5	76	-3.9	104	-1.9

Two patterns emerge. First, low-entropy problems produce $\Delta \approx 0$ pp across all eight conditions: when traces agree, constraints carry no information beyond the majority vote.

Second, the direction of SICA's effect on high-entropy problems depends on the accuracy regime,

not on entropy alone. In the intermediate-SC region (Mistral-7B on FOLIO, default-seed and seed-123 runs), high-entropy Δ reaches $+12.8$ and $+9.1$ pp: traces genuinely disagree, and constraints from correct-answer traces provide discriminative signal that resolves the disagreement. In Type A conditions (Qwen2.5-14B on FOLIO, $\Delta = -8.2$ pp), high-entropy problems represent rare deviations from a strong model's consensus; constraints inherit the model's confirmation bias. In Type C (Mistral-7B on ProofWriter, $\Delta = -4.9$ pp), high-entropy problems reflect systematic errors. The mechanism is not entropy per se but the interaction of entropy with model capability.

O Failure Mode Analysis

The regime-dependent pattern in Table 1 reflects three distinct failure modes of self-extracted logical constraints.

Type A: confirmation bias (SC $\geq$ ~60%). Strong models produce consistent reasoning traces, and constraints extracted from those traces mirror the answer distribution. When most constraints support the same candidate, MAX-SAT excludes few clauses ($|\mathcal{C}^{\setminus}| \approx 0$), the penalty term in Eq. 3 vanishes, and the scoring function degenerates to weighted vote counting (Eq. 4). SICA and SC then produce identical predictions. The ablation in Appendix I confirms this empirically: on Qwen2.5-14B (FOLIO + ProofWriter, $n = 804$), MAX-SAT and count-only baselines agree on 803 of 804 predictions.

Type B: non-formalizable domains. LogiQA problems require pragmatic and contextual reasoning that resists clean logical formalization. Qwen2.5-14B on LogiQA achieves an extraction rate of 86.4%, but the resulting constraints have a contradiction rate of only 0.8%: extracted formulas rarely conflict across traces, leaving MAX-SAT with no logical tension to exploit. Mistral-7B on LogiQA sits in the same SC range (51.00%) yet produces $\Delta = +0.50$ pp ($p = 1.0$); Qwen2.5-14B on the same domain yields $\Delta = -0.50$ pp ($p = 1.0$). Neither model benefits, confirming that the right accuracy regime is necessary but not sufficient when the domain resists formalization.

Type C: error amplification (SC $\leq$ ~40%). When model accuracy is low, the majority of traces reach incorrect answers. Constraints extracted from wrong traces encode incorrect reasoning as formal rules. Z3 treats these rules as soft clauses with weight proportional to their frequency, so the solver preferentially satisfies the wrong constraints and penalizes traces that happen to reach the correct answer. Mistral-7B on ProofWriter ($\Delta = -2.83$ pp, raw $p = 0.014$; BH adj. $p = 0.112$) demonstrates this effect: SICA converts a weak positive signal in trace diversity into a net negative by formalizing the errors.

P Cross-Model Confirmation Bias

Table 16 reports MaxSAT-weighted confirmation bias metrics across three model families on FOLIO ($n = 204$). Weighted bias ratio and wrong-supporting constraint percentage both decrease with model capability, while SICA's accuracy effect shifts from positive (Mistral, intermediate-SC regime) to negative (LLaMA, high-SC regime). Raw bias ratios ($4.97\times$, $3.93\times$, $2.88\times$) are reported in §5.2. LLaMA weighted BR CI: [2.19, 3.28].

Table 16: Cross-model confirmation bias on FOLIO ($n = 204$, $K = 12$). BR (wtd.): MaxSAT-weighted ratio of error-supporting constraints on incorrect vs. correct problems. Wrong %: weighted fraction of constraints supporting the incorrect majority on error problems. Δ: SICA $-$ SC (pp).

Model	BR (wtd.)	Wrong (%)	Δ (pp)
Mistral-7B	$4.24\times$	80.9	$+4.90$
Qwen2.5-14B	$3.69\times$	78.7	-0.98
LLaMA-3.1-8B	$2.66\times$	72.7	-4.41

Frontier model universality. Table 2 (main text, §5.2) extends confirmation bias analysis to nine models spanning 7B open-source to frontier closed-source. All models exhibit BR > 1 (range: 1.66–$4.97\times$). Extraction context varies: open-source models use self-extraction (the model extracts constraints from its own traces); frontier models use cross-model extraction (Mistral-7B extracts FOL from the target model's traces), as closed-source APIs do not expose internal representations suitable for direct symbolic extraction.

ProcessSim definition. Process similarity is the mean pairwise cosine similarity of TF-IDF trace representations:

$$\text{ProcessSim}(q) = \frac{2}{K(K-1)} \sum_{i<j} \cos(\mathbf{v}_i^q, \mathbf{v}_j^q), \tag{7}$$

where $\mathbf{v}_i^q$ is the TF-IDF vector of trace t_i for question q.

Process-κ gradient. Under cross-model extraction (Mistral-7B extracts FOL from other models' traces), SICA gain decreases monotonically with process-level cosine similarity (Table 17; Spearman $\rho = -1.0$, $n = 5$, one-tailed $p = 0.0083$). Gain requires two conditions: diverse traces (low κ_{proc}) *and* substantial self-bias (high BR). R1-Distill-8B satisfies both ($\kappa_{\text{proc}} = 0.248$, BR $= 2.92$) and gains $+2.45$ pp; o3 has moderate trace diversity ($\kappa_{\text{proc}} = 0.596$) but near-minimal bias (BR $= 1.66$, exhibiting a bimodal error pattern: 82% of error problems have BR $= \infty$ but low finite-BR), yielding $\Delta = 0$.

Table 17: Process-κ gradient under cross-model extraction (Mistral-7B $\rightarrow$ target model traces, FOLIO). κ_{proc}: process-level cosine similarity. Δ: SICA $-$ SC (pp). Spearman $\rho(\kappa_{\text{proc}} \rightarrow \Delta) = -1.0$, $p = 0.0083$. *Interim ($n = 95/204$).

Model	κ_{proc}	Δ (pp)	BR
R1-Distill-8B	0.248	+2.45	2.92
gpt-5.5	0.563	+0.49	3.48
o3	0.596	0.00	1.66
Gemini-2.5-Pro*	0.790	−1.05	2.79
gpt-4.1	0.792	−1.47	2.70

Q Independence-Accuracy Scatter Plot

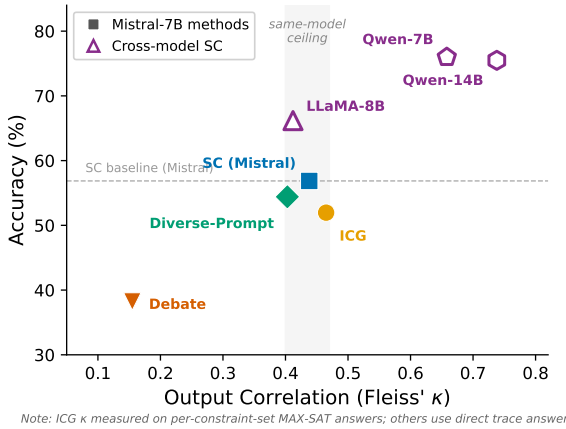


Figure 6: Independence-accuracy trade-off across aggregation methods (Mistral-7B, filled markers) and model families (LLaMA-3.1-8B, Qwen2.5-7B, Qwen2.5-14B; open markers) on FOLIO ($n = 204$). Same-model methods cluster at $\kappa \approx 0.40$–0.47 (gray band), while cross-model SC shows stronger correlations ($\kappa = 0.41$–0.74) at higher accuracy. Debate breaks correlation ($\kappa = 0.15$) at catastrophic accuracy cost. Dashed line: Mistral-7B SC baseline.

Trace independence metrics. Table 19 reports the full 15-condition breakdown of Fleiss' κ and Effective-K at $K = 12$, referenced in §5.1.

Table 18: Independence-accuracy trade-off (FOLIO, $n = 204$). Top: aggregation methods on Mistral-7B; bottom: SC across three model families. All families produce $\kappa \geq 0.41$, confirming an architecture-independent ceiling. κ: Fleiss' kappa on raw answers.

Model	Aggregation	κ	Eff-K	K	Acc. (%)
Mistral-7B	Debate	0.15	4.44	12	38.24
	Diverse-Prompt	0.40	2.21	12	54.41
	Self-Consistency	0.44	2.02	10	56.86
	ICG	0.47	1.93	10	51.96
LLaMA-3.1-8B	Self-Consistency	0.41	2.17	12	66.18
Qwen2.5-7B	Self-Consistency	0.66	1.46	12	75.98
Qwen2.5-14B	Self-Consistency	0.74	1.32	12	75.49

Table 19: Trace independence metrics ($K = 12$, sorted by κ descending). κ: Fleiss' Kappa. K_{eff}: Effective-K (Eq. 1). SC: Self-Consistency baseline. Δ: SICA $-$ SC (pp). $^*p < 0.05$ (uncorrected; BH adj. $p = 0.112$, m=16); positive Δ values do not reach significance. multi-T = $T \in \{0.3, 0.7, 1.2\}$, 4 traces each.

Model	Condition	κ	K_{eff}	SC (%)	Δ (pp)
Qwen3 (think)	FOLIO	0.91	1.09	∼84	∼0
Qwen2.5-14B	LogiQA	0.82	1.20	78.50	−0.50
Qwen3	FOLIO	0.77	1.26	84.80	−0.49
Qwen2.5-14B	FOLIO	0.74	1.30	75.49	−0.98
Qwen2.5-7B	FOLIO	0.66	1.46	75.98	+0.49
Mistral-7B (T=0.3)	FOLIO	0.53	1.77	58.33	−1.47
Mistral-7B	LogiQA	0.51	1.81	51.00	+0.50
Mistral-7B (s456)	FOLIO	0.51	1.82	55.88	0.00
Mistral-7B	FOLIO	0.50	1.84	55.88	+3.43
Mistral-7B (s123)	FOLIO	0.49	1.88	54.41	+2.94
Mistral-7B (s789)	FOLIO	0.41	2.17	54.41	−0.49
Mistral-7B (T=0.5)	FOLIO	0.52	1.80	58.33	−0.98
Mistral-7B (multi-T)	FOLIO	0.48	1.91	57.84	0.00
Mistral-7B (T=1.0)	FOLIO	0.48	1.91	55.39	0.00
Mistral-7B	PW-D5	0.21	3.62	39.33	**−2.83***

R Cross-Model Kappa Analysis

Table 20 compares trace independence across three model families on the same benchmarks. On FOLIO ($n = 204$), all four models produce $\kappa \geq 0.41$: LLaMA-3.1-8B ($\kappa = 0.41$), Mistral-7B ($\kappa = 0.44$), Qwen2.5-7B ($\kappa = 0.66$), and Qwen2.5-14B ($\kappa = 0.74$). The bottleneck is architecture-independent—no family escapes the correlation floor. Within the Qwen family, κ increases from 0.66 (7B) to 0.74 (14B), but Mistral-7B (7B) and LLaMA-3.1-8B (8B) differ by $\Delta\kappa = 0.03$ despite belonging to different families, suggesting that cross-family variation ($\Delta\kappa$ up to 0.22 at similar scale) exceeds within-family scale effects ($\Delta\kappa = 0.08$). Disentangling scale from architecture requires controlled same-family experiments across $\geq$3 scales.

Table 20: Cross-model trace independence ($K = 12$, SC aggregation, FOLIO + ProofWriter). Higher κ indicates stronger correlation and lower effective sample size.

Model	Dataset	κ	K_{eff}	SC (%)
Qwen2.5-14B	FOLIO	0.74	1.32	75.49
Qwen2.5-7B	FOLIO	0.66	1.46	75.98
Mistral-7B	FOLIO	0.44	2.02	56.86
LLaMA-3.1-8B	FOLIO	0.41	2.17	66.18
Qwen2.5-14B	PW-D5	0.62	1.54	—
Mistral-7B	PW-D5	0.21	3.62	39.33

S Cross-Model Process-Level Independence

Table 21 extends the process-level independence analysis (§5.3) to four model families on FOLIO ($n = 204$). Mean pairwise cosine similarity of TF-IDF trace representations exceeds answer-level κ in every family, but the gap is model-strength dependent: it narrows from 0.31 (LLaMA) and 0.23 (Mistral) to 0.02 (Qwen-14B). For weaker models ($r \approx 0$), answer diversity does not reflect reasoning diversity—process-convergent but answer-divergent trace pairs constitute 9.5% of all pairs on Mistral-7B versus 2.5–3.2% on other families. For stronger models ($r \approx 0.23$, $p \leq 0.001$), process and answer agreement align, and answer-level κ already captures the bottleneck. Process-κ thus provides a crucial additional diagnostic specifically for weaker models, where answer-level metrics underestimate reasoning homogeneity.

Table 21: Cross-model process-level independence on FOLIO ($n = 204$, $K = 12$). Mean Cosine: mean pairwise TF-IDF cosine similarity of reasoning traces (Eq. 7). Answer κ: per-question answer agreement. Gap: Cosine $-$ κ. r: Pearson correlation between ProcessSim and answer agreement. Conv.: fraction of process-convergent but answer-divergent trace pairs. ** $p < 0.01$; *** $p < 0.001$.

Model	Mean Cos.	Ans. κ	Gap	r	p	Conv. (%)
Mistral-7B	0.690	0.460	0.230	-0.064	0.362	9.5
LLaMA-3.1-8B	0.759	0.447	0.312	-0.001	0.985	3.2
Qwen2.5-7B	0.790	0.658	0.132	$+0.226$	0.001**	2.5
Qwen2.5-14B	0.766	0.744	0.022	$+0.244$	<0.001***	2.5

Representation robustness. Replacing TF-IDF with SBERT embeddings (all-MiniLM-L6-v2) yields ProcessSim $= 0.91$ (vs. TF-IDF 0.74), widening the gap over answer-level $\kappa = 0.55$. The process homogeneity finding is robust to representation choice.

T Constraint Quality Across Methods

Table 22 compares constraint quality metrics across the three method-level interventions on Mistral-7B FOLIO ($n = 204$). Standard SICA extracts from reasoning traces and achieves the highest parse rate (97.1%) but also the highest redundancy (42.9%): nearly half of all constraints are syntactic duplicates across the 12 traces, quantifying the co-sourcing effect of same-model generation. ICG generates constraints directly from the problem statement without trace conditioning. Its lower parse rate (74.6%) reflects the difficulty of standalone formalization at the 7B scale, though successfully parsed constraints compile at a higher rate (99.4% vs. 96.5%). ICG produces fewer unique constraints per question (24.9 vs. 39.3) with lower redundancy (32.3%), consistent with the absence of trace-level repetition. Contrastive scoring generates pro and con constraint sets for each candidate answer. The format yields far fewer unique constraints per question (9.2) with zero redundancy (each pro/con pair is structurally distinct), but 62.7% of constraints are trivially satisfied by Z3—they impose no logical tension and cannot discriminate between candidates. Only 51.0% of Contrastive constraints are discriminative, compared to 81.4% for standard SICA and 77.0% for ICG.

Table 22: Constraint quality metrics across method-level interventions (Mistral-7B, FOLIO, $n = 204$). Parse: fraction of extraction attempts yielding valid JSON. Z3 Compile: fraction compiling to Z3 formulas. Unique/q: mean unique constraints per question. Redundancy: fraction of constraints duplicated across traces. Disc.: fraction discriminative (different Z3 outcome per candidate). Triv. Sat.: fraction trivially satisfied (no logical tension).

Method	Parse	Z3	Uniq/q	Redund.	Disc.	Triv. Sat.
Standard SICA	97.1%	96.5%	39.3	42.9%	81.4%	—
ICG	74.6%	99.4%	24.9	32.3%	77.0%	—
Contrastive	74.0%	94.4%	9.2	0.0%	51.0%	62.7%

U Error Categorization of Method-Level Interventions

We categorize discordant predictions (where the method and SC disagree) for Debate and Contrastive on FOLIO ($n = 204$) to understand why these interventions degrade accuracy.

Debate discordant analysis. Debate produces 60 correct→wrong flips and 22 wrong→correct flips on FOLIO. Answer-type breakdown reveals that False-labeled questions are most vulnerable:

82.8% of False questions that SC answers correctly are flipped by Debate, compared to 63.9% for True and 25.5% for Unknown. Counterintuitively, failures concentrate not where debate breaks down but where it succeeds most thoroughly: discordant questions average 2.93 effective debate rounds (proposer-critic exchanges producing substantive updates) versus 2.52 for concordant ones. More extensive same-model debate produces more confident but incorrect constraints, consistent with the independence limitation—the critic draws on the same model distribution as the proposer, reinforcing shared biases rather than correcting them.

Contrastive discordant analysis. Contrastive produces 34 discordant questions on FOLIO. All 34 produced non-degenerate scores (both pro and con constraint sets non-empty), compared to 20.1% degenerate scores in the full sample. This indicates that scoring failures on discordant questions stem from systematic pro-bias rather than missing constraints. Among discordant questions, 97.1% contain negation words in the problem statement versus 76.5% in the full sample, suggesting that negation-bearing problems are particularly susceptible to pro-constraint imbalance. The average pro/con score ratio is 1.0 for discordant questions versus 0.68 for the full sample: the method generates balanced-looking scores that mask directional errors.

Cross-method overlap. Of the 79 unique questions on which either method disagrees with SC, only 15 overlap between Debate and Contrastive (Jaccard index $= 0.19$). All 15 shared discordant questions contain negation words. The low overlap confirms that the two methods attack different vulnerability types—Debate through extended same-model argumentation that amplifies confident errors, Contrastive through pro-bias in constraint generation—while sharing a common root cause in correlated same-model voting: both methods rely on the same model's output distribution and cannot introduce genuinely independent signal.

V TOST Equivalence Tests

We apply two one-sided tests (TOST) (Lakens, 2017) with equivalence margin ± 2 pp to all 16 non-degenerate SICA–SC comparisons, classifying each condition as *equivalent* (TOST significant), *genuinely different* (McNemar $p < 0.05$ and TOST not significant), or *underpowered* (neither test significant). Four conditions are equivalent, confirming that SICA and SC produce statistically indistinguishable accuracy. Two conditions show genuine differences, both negative: Mistral-7B ProofWriter (-2.83 pp, $p = 0.014$) and Qwen2.5-14B ProofWriter (-2.50 pp, $p = 0.011$). The remaining ten conditions are underpowered, reflecting the limited statistical power at n=200–204 for detecting $|\Delta| \leq 2$ pp effects.

Margin sensitivity. Table 23 shows that the classification is robust across margins: regardless of the chosen margin, no condition shows a significant positive SICA effect.

Table 23: TOST classification sensitivity across equivalence margins (16 non-degenerate conditions).

Margin	Equivalent	Different	Underpowered
± 1 pp	0	2	14
± 2 pp	4	2	10
± 3 pp	10	0	6

Relative-margin TOST. The absolute ± 2 pp margin is more stringent at high baselines ($\pm 2.4\%$ relative at SC $= 85\%$) than low ones ($\pm 5.1\%$ at SC $= 39\%$). A relative $\pm 5\%$ of SC margin addresses this asymmetry: 8 conditions are equivalent, 2 genuinely different (both negative), and 6 underpowered. The additional four equivalent conditions are all high-SC ($\geq 75\%$), where the wider relative band captures practical indistinguishability. No condition shows a significant positive effect under either margin.

Table 24: Conditional Correction Ratio across eight verifier–generator conditions. CCR > 0: cross-arch NLI on ProofWriter. CCR < 0: same-model SICA or weak capability gap.

Generator–Verifier	Dataset	CCR	Δ (pp)
Mistral × BART-NLI	PW	+1.86	+5.17
Mistral × DeBERTa-NLI	PW	+1.67	+6.67
Mistral × RoBERTa-NLI	PW	+1.42	+4.17
Qwen-14B × DeBERTa-NLI	PW	−0.21	+0.33
Mistral × DeBERTa-NLI	FOLIO	−0.47	+2.94
Mistral × SICA	FOLIO	−2.93	+3.43
Qwen-14B × SICA	PW	−4.21	−2.50
Mistral × SICA	PW	−4.61	−2.83

Conditional Correction Ratio data.

Disagreement-Routed Cascade details. DRC invokes NLI only when SC consensus falls below threshold τ, avoiding unnecessary verifier calls on high-agreement problems. Gains concentrate on weak generators (SC $< 40\%$), consistent with the

capability-gap hypothesis (§6). Regime-dependent bottleneck analysis appears in Appendix X.

Table 25: DRC on Mistral-7B (K=12, DeBERTa-large-mnli, w=3). τ=0.7 recovers 83–90% of full-combo gain at 35–39% NLI cost across both datasets.

Dataset	Method	Acc.	NLI Calls	Cost	Recovery
PW-D5	SC only	39.33%	0/600	0%	—
	DRC τ=0.7	**45.33%**	**233/600**	**38.8%**	**90.0%**
	Full combo	46.00%	600/600	100%	100%
FOLIO	SC only	55.88%	0/204	0%	—
	DRC τ=0.7	**58.33%**	**71/204**	**34.8%**	**83.3%**
	Full combo	58.82%	204/204	100%	100%

W Multi-Verifier NLI Results

Table 26: Cross-architecture NLI combo on ProofWriter depth-5 (n=600, Mistral-7B traces, SC = 39.33%). Three NLI verifiers × three weight settings. $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (McNemar).

Verifier	Standalone	w	Δ (pp)	p
DeBERTa-MNLI	55.00%	1	+0.83	n.s.
		3	+6.67	<0.001***
		5	+11.00	<0.001***
RoBERTa-MNLI	37.67%	1	−0.17	n.s.
		3	+4.17	0.003**
		5	+4.33	0.037*
BART-MNLI	46.17%	1	0.00	n.s.
		3	+5.17	<0.001***
		5	+6.33	0.003**

All verifiers fine-tuned on MultiNLI. Unknown recall <3.5% across all verifiers.

Three-verifier BH correction. Table 27 applies Benjamini–Hochberg correction across all 15 Mistral-7B verifier–weight conditions (m=15, α=0.05; distinct from the m=12 cross-generator analysis in §6). Ten of 15 conditions remain significant after correction, confirming that multi-verifier NLI gains are robust to multiple testing.

Table 27: BH correction across 3 verifiers × 5 weights on Mistral-7B ProofWriter (n=600, m=15). ✓: $p_{\text{BH}} < 0.05$.

Verifier	w=1	w=2	w=3	w=4	w=5
DeBERTa-MNLI		✓	✓	✓	✓
RoBERTa-MNLI			✓	✓	✓
BART-MNLI			✓	✓	✓

10/15 significant at α=0.05 after BH correction. All w=1 conditions and RoBERTa w=2 are non-significant.

X Capability Gap Data Points

Table 3 reports the controlled cross-generator comparison with DeBERTa-large-mnli on ProofWriter (n=600). We extend this analysis by pairing Mistral-7B with two additional NLI architectures: BART-large-mnli (46.17% standalone, combo +5.17 pp, p = 0.0002) and RoBERTa-large-mnli (37.67% standalone, combo +4.17 pp, p = 0.003). All three NLI architectures produce significant gains on Mistral-7B at $w \geq 3$.

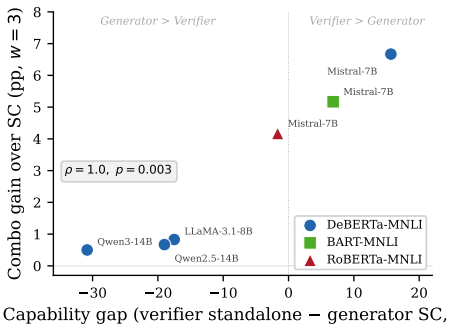


Figure 7: Combo gain vs. capability gap (verifier standalone − generator SC) at w=3 across three NLI architectures. Gain decreases monotonically (Spearman ρ = 1.0, exact permutation p = 0.003, n=6; four of six conditions share the DeBERTa verifier—see Limitations); significant gains occur only when the verifier exceeds the generator.

Across all six ProofWriter verifier-generator pairs at w=3 (three verifiers × Mistral-7B plus three negative-gap generators × DeBERTa), combo gain decreases monotonically with the capability gap (Spearman ρ = 1.0, exact permutation p = 0.003, n = 6; however, four of six conditions share the DeBERTa verifier, which may reduce effective degrees of freedom—see Limitations; Figure 7). Only conditions where the verifier's standalone accuracy exceeds the generator's SC accuracy show significant improvement, confirming Finding 2. After BH correction (m=12), only Mistral-7B w=3 (adj. $p = 6.78 \times 10^{-5}$) and w=5 (adj. $p = 7.41 \times 10^{-6}$) survive.

Robustness check. Including all nine conditions (three verifiers × three weight levels on Mistral-7B) yields Spearman ρ = 0.836 (exact permutation p = 0.007, n = 9), confirming that the monotonic trend is robust to the choice of weight parameter.

Full NLI combo with BH correction.

Generator×Extractor 2×2 Matrix. Table 29 reports the full 2×2 generator×extractor matrix on FOLIO (n=204). The generator dominates accuracy (row difference ∼25 pp vs. column difference ∼3 pp), contrasting with ProofWriter where the extractor dominates (Δ range ±40–50 pp). Cross-extractor McNemar tests: Mistral-7B gen, Δ = +2.94 pp, p = 0.143 (n.s.); Qwen3-14B gen, Δ = +2.94 pp, p = 0.021 (statistically significant but practically small: 6/204 problems changed). The

Table 28: Full cross-generator NLI combo results on ProofWriter depth-5 (n=600) with Benjamini–Hochberg correction (m=12, α=0.05). All conditions use DeBERTa-large-mnli verifier (55.00% standalone). Two of twelve conditions survive correction (both Mistral-7B). * $p_{\text{BH}} < 0.05$; *** $p_{\text{BH}} < 0.001$.

Generator	w	Combo (%)	Δ (pp)	p_{raw}	p_{BH}
Mistral-7B SC 39.33%, gap +15.67	1	40.17	+0.83	.3593	.7208
	3	46.00	+6.67	1.1×10^{-5}	6.78×10^{-5} ***
	5	50.33	+11.00	6.2×10^{-7}	7.41×10^{-6} ***
LLaMA-3.1-8B SC 72.50%, gap −17.50	1	73.33	+0.83	.3593	.7208
	3	73.33	+0.83	.6646	.8862
	5	72.00	−0.50	.8708	1.0
Qwen2.5-14B SC 74.00%, gap −19.00	1	74.17	+0.17	1.0	1.0
	3	74.67	+0.67	.5034	.7552
	5	75.17	+1.17	.3604	.7208
Qwen3-14B SC 85.83%, gap −30.83	1	86.00	+0.17	1.0	1.0
	3	86.33	+0.50	.4531	.7552
	5	87.00	+1.17	.2100	.7208

identical net Δ is coincidental—the two generators have different discordant-pair distributions that happen to yield the same net advantage. BR > 1 in all eight cells across both datasets (range 1.94–2.70), confirming SCB universality under cross-model extraction.

Table 29: FOLIO 2×2 generator×extractor matrix (n=204). SC: Self-Consistency baseline. κ: answer-level Fleiss' kappa. BR: bias ratio.

Generator	Extractor	SC (%)	SICA (%)	Δ (pp)	BR
Mistral-7B κ=0.46	Mistral	55.88	56.86	+0.98	2.70
	Qwen3	55.88	59.80	+3.92	1.94
Qwen3-14B κ=0.79	Qwen3	83.33	84.80	+1.47	2.35
	Mistral	83.33	81.86	−1.47	2.66

Y Single-Trace and Aggregation Statistics

Table 30: Single-trace accuracy (mean ± std over K=12 traces) and SC@12 majority vote on FOLIO (n=204). Aggregation gain = SC@12 − single-trace mean. SC@12 uses alphabetical tiebreaking (consistent with Table 1).

Model	Single (%)	Std	SC@12 (%)	Gain (pp)
Mistral-7B	46.41	2.80	55.88	+9.47
Qwen3-14B	79.37	1.98	84.80	+5.43
Qwen3-14B (think)	85.42	0.99	86.27	+0.86

Inter-trace standard deviation shrinks 2.8× from Mistral-7B to Qwen3-14B thinking mode, reflecting the diversity compression that underlies the aggregation ceiling (§6).

Extraction temperature sensitivity. Controlling for the trace set (excluding $T_{\text{ext}} = 0.3$, which uses a different seed), all conditions yield $\Delta \leq +1.47$ pp (n.s.), indicating that extraction temperature has no significant effect on SICA accuracy. This supports the structural nature of symbolic confirmation bias: the bottleneck is what gets extracted, not the temperature at which extraction occurs.

Table 31: Effect of extraction temperature T_{ext} on SICA accuracy (Mistral-7B, FOLIO n=204). † Uses a different trace set (SC = 54.41%); all other conditions share the same traces (SC = 55.88%).

T_{ext}	SICA (%)	Δ vs. SC (pp)	Avg. #Constr.
0.1	57.35	+1.47	69.3
0.3†	59.31	+3.43	68.8
0.5	56.37	+0.49	68.2
0.7 ***	53.43	−2.45	67.3
1.0 ***	55.39	−0.49	65.2

Table 32: Multi-seed replication results. Each row is an independent seed with fresh traces (K=12, T=0.7). Mean ± SD computed across seeds. Mistral-7B shows directionally unstable Δ (+0.16 ± 2.70 pp); all Qwen2.5-14B seeds show $\Delta \leq 0$.

Seed	SC (%)	SICA (%)	Δ (pp)
Mistral-7B × FOLIO (n=204)[a]			
123	54.41	57.35	+2.94
456	55.88	55.88	0.00
789	59.31	56.86	−2.45
Mean ± SD	56.53 ± 2.51	56.70 ± 0.75	+0.16 ± 2.70
Qwen2.5-14B × FOLIO (n=204)			
42	77.50	75.49	−2.01
123	74.51	73.04	−1.47
456	74.02	74.02	0.00
789	74.02	72.55	−1.47
Mean ± SD	75.01 ± 1.67	73.78 ± 1.30	−1.24 ± 0.86

[a]Original baseline (exp033) used no fixed seed and is excluded from seed statistics.

Multi-seed replication.

Y.1 Sub-population Analysis

Table 33: SICA vs. SC accuracy by gold-label type (Mistral-7B). Corrected SC baseline uses alphabetical tiebreaking. Δ pp = SICA − SC. p-values from exact McNemar test (two-sided).

Dataset	Label	n	SC	SICA	Δ pp	p
FOLIO-204	True	72	56.9	62.5	+5.6	.125
	False	63	42.9	42.9	±0.0	1.0
	Unknown	69	66.7	71.0	+4.3	.250
	Overall	204	55.9	59.3	+3.4	.092
PW-600	True	200	8.5	6.0	−2.5	.063
	False	200	19.5	14.5	−5.0	.076
	Unknown	200	90.0	89.0	−1.0	.774
	Overall	600	39.3	36.5	−2.8	.014*

On FOLIO-204 with the corrected SC baseline (alphabetical tiebreaking), SICA improves over SC by +3.4 pp overall (55.9% → 59.3%), though the aggregate McNemar test is marginally non-significant (p=.092). The gain concentrates in the mid-constraint tertile (+9.4 pp, p=.031), where sufficient symbolic evidence exists for MaxSAT to

Table 34: SICA vs. SC accuracy by extracted constraint count tertile (Mistral-7B). Corrected SC baseline (alphabetical tiebreaking). Constraint counts are unique deduplicated constraints per problem.

Dataset	Tertile	n	$\bar{c}$	SC	SICA	Δpp	p
FOLIO-204	Low [0–32]	74	21.7	62.2	63.5	+1.4	1.0
	Mid [33–47]	64	39.7	50.0	59.4	+9.4	.031*
	High [48–83]	66	58.7	54.5	54.5	±0.0	1.0
PW-600	Low [0–39]	212	28.8	43.9	39.6	−4.3	.078
	Mid [40–52]	196	45.8	40.8	37.2	−3.6	.143
	High [53–99]	192	63.7	32.8	32.3	−0.5	1.0

meaningfully re-rank answers; in the low- and high-constraint groups the methods perform comparably. On ProofWriter-600, SICA shows a small uniform degradation consistent with the main results, attenuating in the high-constraint tertile ($\Delta{=}-0.5$ pp, $p{=}1.0$). We note that with six subgroup tests, one significant result at $\alpha{=}0.05$ is expected by chance; the mid-constraint finding should be interpreted as exploratory.

Individual intervention significance. Of the 27 training-free remediation strategies (Figure 2), the two largest positive effects are Cross-model Qwen3→Mistral (+0.98 pp) and Generational (+0.49 pp); neither approaches statistical significance ($p > 0.5$ for both). The largest negative effects are LLM-Debate (−18.6 pp, $p < 0.001$) and Contrastive Scoring (−9.3 pp, $p < 0.001$). After BH correction across all 27 comparisons, only these two negative results remain significant. No strategy produces a significant positive effect under any correction procedure.

Z Extended Discussion and Limitations

SCB versus CoT unfaithfulness. SCB is distinct from CoT unfaithfulness (Turpin et al., 2024) and CoT-level confirmation bias (Wan et al., 2025; Jhaveri et al., 2026). CoT unfaithfulness concerns whether stated reasoning reflects actual computation; SCB concerns whether *self-extracted formal constraints* mirror the model's answer distribution. Even if traces were perfectly faithful, self-extraction would still inherit the distributional bias—the constraint $P(\text{answer_A})$ reflects vote share, not logical validity. The shuffle control (§5.2) confirms this: randomizing constraint–answer associations preserves BR, showing the bias is in the trace-count distribution, not in extraction-specific errors. However, if traces are unfaithful, the mirage may be worse than our measurements suggest, because constraints would reflect not the model's reasoning but its post-hoc rationalization.

Implications for neuro-symbolic practitioners. Our results do not imply that neuro-symbolic methods are inherently flawed—they imply that *self-extracted* constraints cannot provide independent verification. The key distinction is *when* formalization occurs relative to answer generation. Specification-generation approaches (e.g., SatLM (Ye et al., 2024), Logic-LM (Pan et al., 2023)) formalize the problem *before* generating candidate answers, avoiding the extraction bottleneck; SatLM's success confirms that the formal verification mechanism is sound when constraints are independently derived. Trace-extraction (this work) operates post-hoc on reasoning traces that already contain candidates, inheriting the distributional prior that generates SCB. When post-hoc verification is unavoidable, practitioners should employ external verifiers from a different architecture family (§6) and use the diagnostic framework in §7 to assess feasibility before deployment. The SCB diagnostic toolkit (below) provides detailed computation methods and threshold recommendations.

Test-time scaling implications. Vote diversity decreases with model capability: K_{eff} drops from 3.62 (Mistral-7B on ProofWriter) to 1.09 (Qwen3-14B thinking mode). This creates a fundamental tension for o1/R1-style scaling (Guo et al., 2025): longer reasoning chains produce more coherent but less diverse traces, reducing the marginal value of additional samples. Our Process-κ analysis (§5.3) reveals that this homogenization operates at the reasoning-process level, not just answer level—traces that disagree on final answers still share 69% of their reasoning structure. Cross-architecture ensembles offer a path forward, but the capability-gap condition (§6) implies that the verifier must be genuinely complementary, not just different.

SCB diagnostic toolkit. We recommend four zero-cost diagnostics, computable from sampled traces alone, to predict whether symbolic self-verification will help or hurt before deployment. *Bias Ratio* (BR; §5.2): for each problem, compute the fraction of constraints supporting the majority answer, divided by majority vote share. BR > 1 indicates constraints are biased toward the majority; BR $\gg 2$ (as in all nine models tested; Table 2) signals that extraction amplifies rather than checks the model's prior. Report as mean ± bootstrap 95% CI. *Effective-K* (K_{eff}; Eq. 1): compute Fleiss' κ

across K traces, then $K_{\text{eff}} = 1 + (K-1)(1-\kappa)$. Threshold: $K_{\text{eff}} \leq 1.3$ never improves in our experiments—aggregation collapses to majority voting. $K_{\text{eff}} > 1.8$ is a necessary (not sufficient) condition for gain. *Degenerate Consensus* (C_0; §3.1): the fraction of problems where all K traces agree. $C_0 > 95\%$ renders aggregation moot; $C_0 < 5\%$ means the model is maximally uncertain and constraints tend to amplify noise. *ProcessSim* (Eq. 7): mean pairwise cosine similarity of TF-IDF trace vectors. ProcessSim > 0.6 indicates homogeneous reasoning; combined with BR > 2, this predicts SCB will dominate. A practitioner should compute all four on a held-out sample ($n \geq 50$). If $K_{\text{eff}} \leq 1.3$ or $C_0 > 95\%$ or (BR > 2 and ProcessSim > 0.6), symbolic self-verification is unlikely to help.

Future directions: trained extractors and beyond. Our oracle controls reveal that perfect extraction would yield $+29.90$ pp over SC on Mistral-7B FOLIO—self-extraction captures only $\sim$12% of this ceiling. This gap motivates several future directions. *Trained constraint extractors*: fine-tuning a small model (e.g., encoder-only) on human-annotated FOL from logic benchmarks could produce genuinely discriminative constraints that break the distributional prior. The key design constraint is structural independence—the extractor should not share training data or architecture with the generator, mirroring the cross-architecture NLI escape (§6). *Multi-model ensembles*: rather than single cross-architecture pairs, combining M structurally independent verifiers could further reduce correlated errors. The capability-gap condition (§6) predicts that each verifier must exceed the generator on the target task to contribute positively; empirical verification across heterogeneous verifier pools is needed. *External knowledge grounding*: connecting to knowledge bases (Wikidata, domain ontologies) could inject verification signal absent from any model's parameters. This avoids SCB entirely by grounding constraints in external facts rather than self-generated traces. *SCB-aware aggregation*: current MAX-SAT aggregation treats all constraints equally. Weighting constraints by estimated independence (e.g., inverse ProcessSim with other traces' constraints) or by discriminative power (e.g., cross-validation on held-out traces) could partially mitigate SCB without requiring external verifiers.

SCB as an evaluation framework. The SCB metrics—BR, K_{eff}, ProcessSim, and CCR—can serve as a general evaluation framework for self-improvement methods beyond SICA. Any method that extracts structured representations from a model's own outputs (self-verification (Huang et al., 2024), self-refinement (Madaan et al., 2023), chain-of-verification (Dhuliawala et al., 2023), self-consistency (Wang et al., 2023)) faces the distributional prior that generates SCB. We propose that future work on self-improvement methods report: (1) BR on error cases (does the method's verification signal correlate with the model's majority answer, or does it provide independent information?); (2) K_{eff} or an analogous diversity metric (does the method generate genuinely diverse candidates, or does it amplify a single mode?); (3) CCR (does the method preferentially correct errors, or does it preferentially confirm them?). A method with BR < 1, high K_{eff}, and CCR > 0 demonstrates genuine self-improvement; BR > 1 with CCR < 0 indicates that the method suffers from SCB regardless of aggregate accuracy gains, which may reflect favorable base rates rather than corrective signal.

Extended limitations. Beyond the limitations discussed in the main text, several additional constraints bound our conclusions. Seven of our eleven model-domain conditions have $n \leq 60$, providing insufficient statistical power for confirmatory claims; power analysis shows that the minimum detectable effect exceeds 5 pp for these conditions (Table 6), so they serve as regime characterization rather than definitive tests. Our extraction prompts use a single zero-shot template per domain; alternative prompting strategies (few-shot exemplars, chain-of-thought extraction, or iterative refinement) might yield higher-quality constraints, though the oracle gap suggests the bottleneck is structural rather than prompt-specific. The constraint quality analysis (91.8% discriminative rate, 39.2% source-trace minority weight) covers Qwen2.5-14B on FOLIO + ProofWriter ($N = 804$); other model-domain combinations may exhibit different characteristics. Constraint extraction uses $T_{\text{ext}} = 0.3$ while trace sampling uses $T = 0.7$; alternative extraction temperatures might yield different results. Although we prompt for first-order logic formalizations, 99.0% of extracted constraints are propositional at the Z3 level, with only 1.0% using true quantifiers ($\forall/\exists$); whether models capable of genuine quantified reasoning would yield higher

discriminative rates is untested. Deduplication uses syntactic equivalence (global average dedup ratio: 46.0%), so semantically equivalent constraints with different surface forms receive separate weights; Z3-based semantic deduplication removes an additional $\sim$10% of constraints and yields a small but significant improvement over no deduplication ($p = 0.013$), suggesting room for better dedup strategies. The GSM8K pilot ($n = 30$) illustrates the constraint-absent regime and should not be interpreted as confirmatory evidence.